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SURVEY

A Review of Machine Learning and Deep Learning Trends in EEG-Based Epileptic Seizure Prediction

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ABSTRACT Epileptic seizures impair patients' health and quality of life, and electroencephalography (EEG)-based prediction enables timely intervention. Early work on epileptic seizure prediction employed linear models, whereas recent studies employ more advanced machine learning and deep learning models. We review 149 studies (2018—May 2025), summarizing trends and gaps across the full pipeline in EEG-based epileptic seizure prediction. We find that four persistent challenges related to this domain remain. First, severe dataset imbalance, with overreliance on Children's Hospital Boston–Massachusetts Institute of Technology EEG dataset (CHB-MIT), minimal use of intracranial electroencephalography (iEEG) datasets, and infrequent integration of public and private datasets. Second, preprocessing practices remain rigid, often involving segmentation into windows of 10 s or less, fixed preictal horizons of 30 to 60 min, basic filtering and undersampling procedures. Third, limited model diversity, as most studies use convolutional neural networks (CNNs), with scant exploration of Transformers or graph neural networks (GNNs), dimensionality reduction is predominantly implicit and few linear baselines. Fourth, validation and deployment are fragmented, with fixed warning horizons, single evaluation protocols, missing clinical metrics, static thresholds or k-of-n voting, minimal interpretability, and few real-world implementations. Future work should assemble heterogeneous EEG datasets, adopt adaptive multiscale preprocessing, and build unified benchmarks that pit CNN, Transformer, GNN, and related models against well-tuned linear baselines—advancing flexible non-convolutional designs beyond the time-frequency paradigm, mandating ablations that contrast implicit with explicit, task-aware dimensionality reduction, and pairing these evaluations with clinically aligned validation, dynamic risk-aware thresholds, embedded interpretability, and prospective testing on energy-efficient edge hardware in ward and home settings.

INDEX TERMS Electroencephalography, epileptic seizure prediction, machine learning, deep learning.

I. INTRODUCTION

Epilepsy is a major neurological disorder affecting approximately 50 million people worldwide. It constitutes a significant cause of morbidity, disability, and mortality, particularly in low- and middle-income countries where access to specialized care is limited [1], [2]. Despite advances in pharmacological and surgical treatments, nearly one-third of patients with epilepsy still experience drug-resistant seizures [3]. In addition to the direct health impacts, epilepsy imposes substantial socioeconomic burdens, including healthcare costs, employment challenges, and diminished quality of life for patients and their families.

Epileptic seizures are abrupt manifestations of abnormal, excessive neuronal activity in the brain, which can cause serious injuries, accidents, and sudden unexpected death in epilepsy [4]. Beyond these immediate risks, recurrent seizures also impose substantial psychological burdens that

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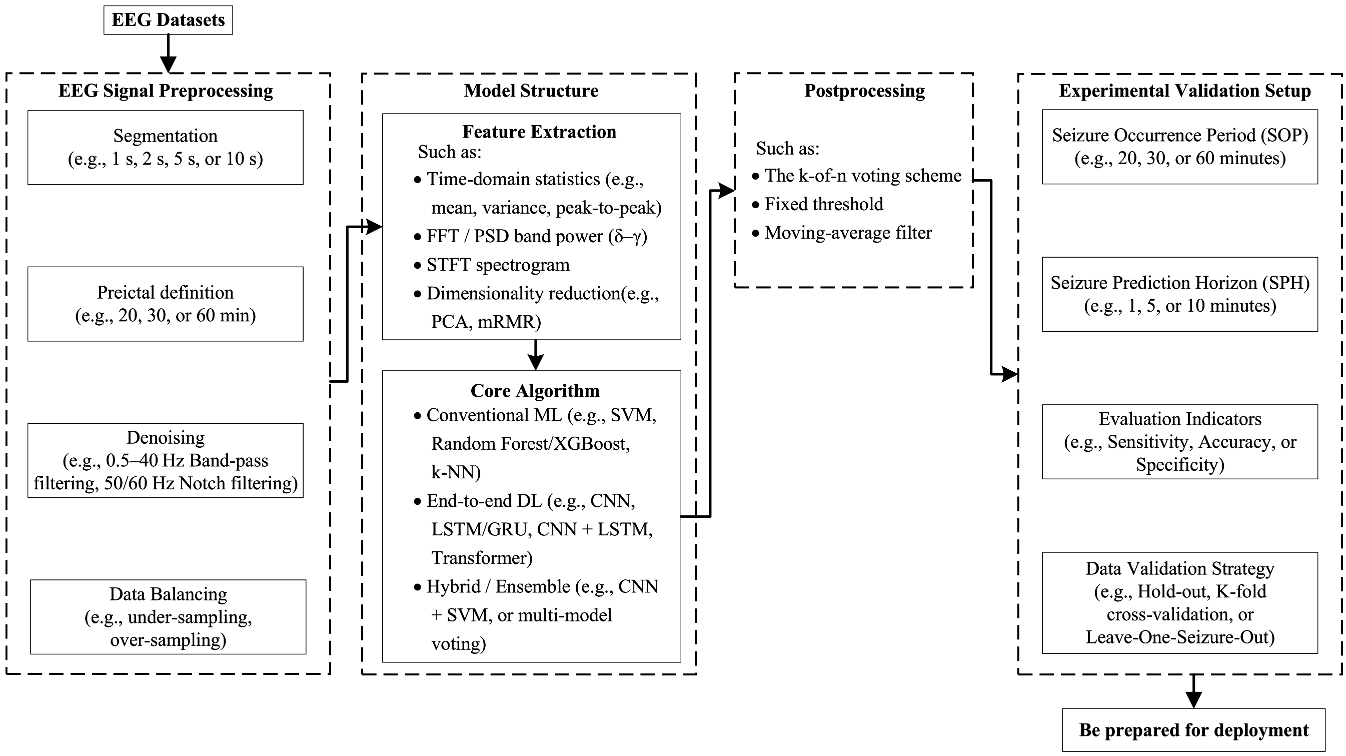

FIGURE 1. **Flowchart of the general process for EEG-based seizure prediction.**

severely impair quality of life. In epilepsy management, monitoring systems fall into two categories: seizure detection and seizure prediction. Detection systems identify ictal events at onset or within seconds. They are evaluated by sensitivity, specificity, false detection rate per hour (FDR/h), and detection latency to enable real-time interventions such as closed-loop neurostimulation or caregiver alerts [5]. In contrast, prediction systems aim to forecast seizures during the preictal period with horizons of 60, 90, or 120 min; their performance is measured by sensitivity, false prediction rate per hour (FPR/h), and mean warning time to provide sufficient lead time for preventive action. Because uncontrolled seizures have severe consequences, effective prediction has become a critical goal in epilepsy care [6]. By issuing warnings, prediction systems allow patients to reach safe environments, initiate rescue therapies, or alert caregivers, thereby reducing morbidity and mortality and improving overall quality of life.

The core idea of seizure prediction is that there exists a discernible transition period (the preictal state) whose timely detection allows for warning of an impending event. Mormann et al. [7] systematically argued that, to surpass random chance truly, models must be patient-specific, have clearly defined prediction horizons, and be validated with rigorous statistical tests. Among various physiological modalities explored for seizure forecasting, electroencephalography (EEG) has emerged as the primary and most clinically validated tool [8]. EEG provides direct insights into the brain's electrical activity, capturing subtle preictal changes that precede seizure onset, and plays a central role in both seizure prediction research and clinical applications.

EEG offers several unique advantages for seizure prediction, including millisecond-scale temporal resolution that is critical for detecting the rapid electrophysiological transitions preceding seizures [9]. Unlike neuroimaging modalities such as magnetic resonance imaging (MRI) or computed tomography (CT), EEG enables continuous, real-time monitoring of brain activity, making it particularly well suited for dynamic seizure forecasting [10]. In clinical practice, EEG not only supports the diagnosis of epilepsy but also contributes to seizure classification, epilepsy subtype identification, treatment monitoring, and presurgical evaluation [10], [11]. Nevertheless, conventional EEG interpretation remains manual and subjective, with reported sensitivity after a first clinical seizure ranging between 29% and 55% [11]. The complexity, variability, and volume of EEG data necessitate automated analysis techniques to improve diagnostic accuracy, scalability, and reliability, particularly in long-term monitoring settings.

Early EEG-based studies often relied on simple linear classifiers—such as linear discriminant analysis (LDA), least-squares linear discriminant (LSLD), and logistic regression (LR)—which provided crucial insights into the temporal electrophysiological signatures preceding seizures [12], [13]. Nonetheless, because these models assume a single linear decision boundary, they struggle to capture the nonlinear, high-dimensional dynamics of multichannel EEG signals, limiting their ability to model the subtle spatiotemporal interactions that foreshadow imminent seizures. To transcend these limitations, researchers have progressively adopted artificial intelligence (AI) approaches, beginning with machine learning pipelines and more recently embracing deep learning architectures that can autonomously learn the intricate nonlinear spatiotemporal patterns embedded in EEG recordings [14], [15]. Alotaiby et al. [5] applied subject-specific typical spatial pattern filters followed by machine learning to scalp EEG recordings, achieving an average sensitivity of 0.89, 0.39 false alarms per hour, and a mean warning time of approximately 69 min. Bandarabadi et al. [16] showed that the optimal preictal window varies across seizures and introduced an amplitude-distribution-histogram overlap criterion to select this window automatically, thereby improving class separability and providing an interpretable predictability metric. Recent AI pipelines have demonstrated higher sensitivity, lower false-positive rates, and greater reproducibility than conventional expert visual inspection or simple heuristic postprocessing rules [17]. Reviews suggest that AI-assisted seizure prediction models can achieve experimental accuracies exceeding 80%; however, real-world deployment remains challenging because of dataset heterogeneity and limited model interpretability [15], [17].

As shown in Fig. 1, the general workflow of EEG-based seizure prediction begins with preprocessing of the raw recordings, including signal segmentation, preictal window definition, noise removal, and balancing preictal and interictal samples. Next, feature extraction produces informative descriptors from the time, frequency, and time–frequency domains, among others. These features are input to a prediction model, whose outputs can be refined by postprocessing routines to enhance performance. Finally, the complete system undergoes experimental validation and is prepared for deployment.

This paper aims to provide a comprehensive overview of the development and evolving trends in machine learning and deep learning approaches for EEG-based epileptic seizure prediction. Several prior systematic reviews have addressed the application of machine learning and deep learning in epilepsy research, but each exhibits important gaps. Shafiezadeh et al. [18] focused solely on cross-patient seizure prediction, neglecting patient-specific modeling and complete coverage of the end-to-end pipeline. Ong et al. [19] reviewed medical devices for epilepsy management, emphasizing hardware development while giving limited attention to algorithmic strategies. Saeidi et al. [20] offered a broad survey of EEG neural decoding, in which seizure prediction played only a peripheral role. More recently, Tautan et al. [21] examined unsupervised learning methods but did not provide an extensive treatment of mainstream supervised and deep learning approaches. Wong et al. [22] evaluated publicly available EEG datasets for seizure detection and prediction, yielding valuable insights into dataset characteristics but stopping short of a comprehensive analysis of algorithmic methodologies or frameworks for clinical translation.

TABLE 1. Inclusion and exclusion criteria for study selection.

Criteria Type	Criteria Description
Inclusion Criteria	
I1	Studies that utilize EEG signals
I2	Studies that focus on epileptic seizure prediction
I3	Studies that employ machine learning or deep learning techniques
Exclusion Criteria	
E1	Studies that were published before 2018
E2	Studies that were not written in English
E3	Studies that did not involve human subjects
E4	Studies that were categorized as review articles

Although these prior reviews offer valuable insights, they remain largely narrow in scope, each addressing only a specific subset of topics within the broader domain of epileptic seizure prediction. None delivers an integrated, systematic, and comprehensive treatment that simultaneously encompasses methodological approaches, dataset utilization, evaluation frameworks, and clinical applicability. Consequently, a critical gap persists: there is currently no review that holistically maps the whole landscape of machine learning and deep learning applications to EEG-based seizure prediction. To address this unmet need, we conducted a comprehensive review with the following objectives:

(i) to identify and synthesize current trends and advancements in seizure prediction research using EEG data;
(ii) to highlight methodological limitations present in current seizure prediction studies;
(iii) to propose future research directions and strategies to bridge the gap between experimental development and real-world clinical implementation.

The structure of this review is as follows: Section II outlines the review methodology, including the search strategy, selection criteria, and data extraction process. Section III summarizes key findings by comparatively analyzing datasets, EEG preprocessing, model architectures, validation setups, postprocessing strategies, and clinical interpretability and deployment. Section IV discusses key trends, strengths, limitations, and challenges in the field. Section V concludes the review and proposes directions for future research.

II. MATERIALS AND METHODS

To promote transparent reporting and facilitate retrieval of studies relevant to its objectives, this review followed established best-practice guidelines, and data were collected as follows. Two primary search queries—"EEG machine learning epilepsy prediction" and "EEG deep learning epilepsy prediction"—were employed to capture the core themes. Literature searches were conducted in the *Web of*

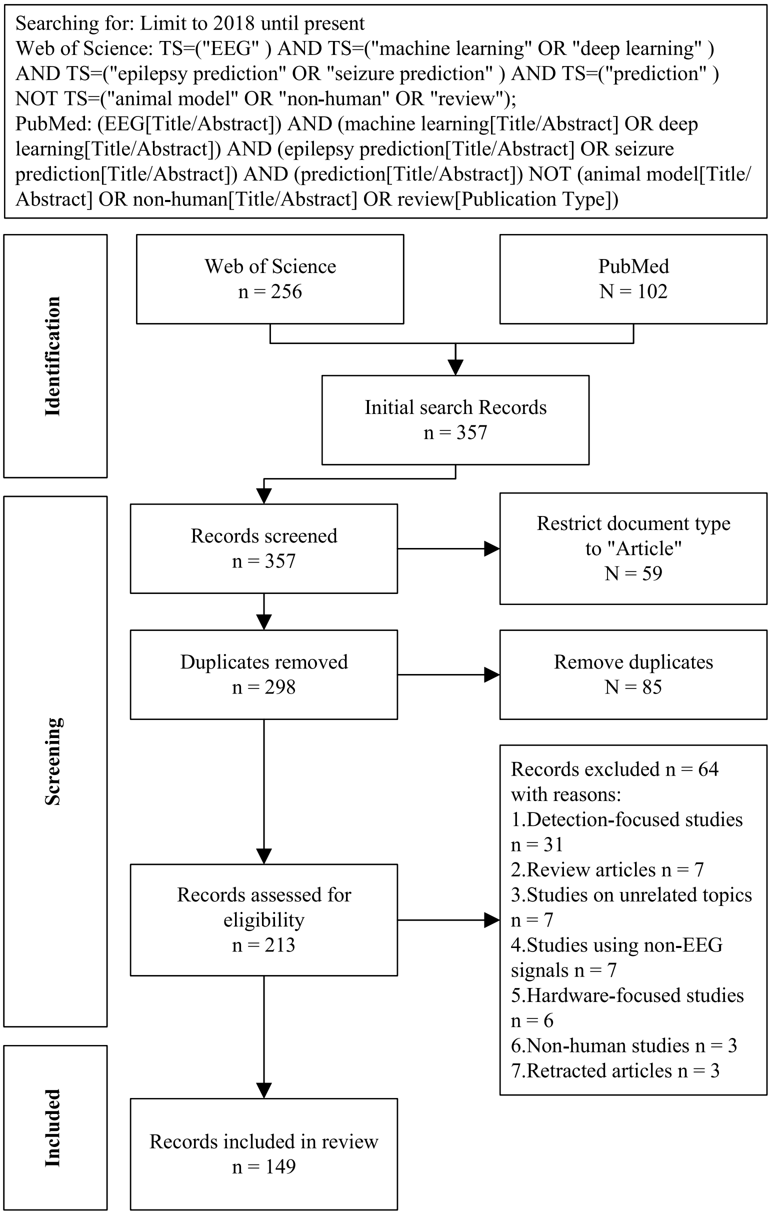

FIGURE 2. **Flowchart of literature search, screening, and selection process for EEG-based epileptic seizure prediction studies.**

Science and *PubMed* databases, and study selection was governed by predefined inclusion and exclusion criteria (Table 1).

Based on the inclusion and exclusion criteria summarized in Table 1, a comprehensive search was conducted to identify the latest research published since 2018. The electronic search was executed on 1 May 2025. Although studies published prior to 2018 were relatively sparse, recent years have seen a pronounced surge in research activity, reflecting substantial methodological and technological evolution. The initial retrieval yielded 256 records from the *Web of Science* and 102 from *PubMed*, for a combined total of 357 records.

To provide an overview of the literature screening procedure, a summary of the search and selection process is illustrated in Fig. 2. An initial screening was conducted using the built-in database filters to restrict the document type to "Article". Duplicate entries were then identified and removed with *Zotero*, leaving 213 unique records for further evaluation.

Only records that passed the initial title and abstract screening were retrieved in full text and subjected to further assessment against the predefined inclusion criteria. Full-text evaluation involved a comprehensive review of each record's title, abstract, and entire content. At this stage, the following records were excluded:

- 31 records that focused primarily on epileptic seizure detection rather than prediction,
- Seven records whose full texts revealed that they were review-type studies,

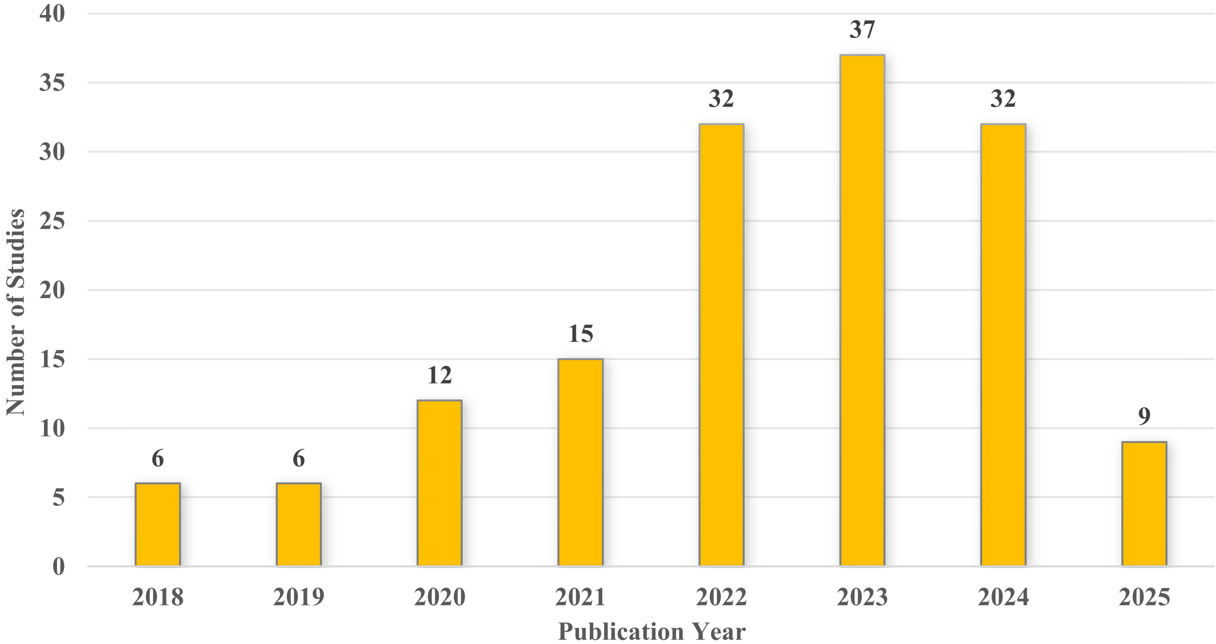

FIGURE 3. Number of included studies by year (from 2018 till 1 May 2025).

- Seven records addressing non-standard topics (e.g., traumatic brain injury prognosis, postoperative outcomes, neonatal hypoxic–ischemic encephalopathy, aura detection, simulations, cognitive monitoring),
- Seven records that used non-EEG signals (e.g., electrocardiography, questionnaires, functional near-infrared spectroscopy),
- Six records that focused primarily on hardware development,
- Three records that involved non-human subjects,
- Three retracted articles.

After the application of all predefined inclusion and exclusion criteria [23], a total of 149 studies were deemed eligible and included for qualitative synthesis. These studies form the basis of the analysis presented in this review.

III. RESULTS

From 2018 through 2025, research on EEG-based epileptic seizure prediction using machine learning and deep learning has shown a steady upward trend, with 2025 data covering publications up to May 1. As illustrated in Fig. 3, only six studies were published annually in both 2018 and 2019, reflecting the early stage of methodological exploration in this domain. The number of studies increased to 12 in 2020 and 15 in 2021, indicating growing research momentum. A sharp rise was observed in 2022 and 2023, with 32 and 37 studies published, respectively—marking the peak of research activity. Although there was a slight decline to 32 studies in 2024, the field remained highly active. For 2025, only nine studies were included at the time of this review, as the year was still in progress. This overall trend underscores the increasing academic and clinical interest in applying machine learning/deep learning methods to EEG-based seizure prediction in recent years.

Among the 149 studies on seizure prediction, deep learning techniques were employed in 107 studies, reaffirming their role as the dominant approach in the field. Traditional machine learning methods were utilized in 22 studies. In addition, researchers have begun to explore integrated strategies: 11 studies proposed hybrid models that combine deep learning and machine learning, while another nine studies conducted explicit comparisons between the two paradigms to benchmark their respective strengths and limitations.

This section provides a comprehensive analysis of the 149 studies included in this review, structured around the key methodological components identified across the literature. Section III-A outlines the datasets utilized for EEG-based seizure prediction. Section III-B reviews the EEG signal preprocessing techniques applied prior to model development. Section III-C describes the model architectures employed for seizure forecasting. Section III-D summarizes the experimental validation setups, including data partitioning strategies and evaluation metrics. Section III-E discusses postprocessing decision mechanisms designed to refine predictive outputs. Section III-F addresses issues related to

TABLE 2. Overview of datasets used in EEG-based seizure prediction studies.

Dataset Name	Open Access	EEG Type	Sampling Rate (Hz)	Channels	Freq.	Ref.
CHB-MIT	yes	sEEG	256	23	107	[24]–[130]
AES-Kaggle	yes	iEEG	500,5000	16	24	[85]–[101], [112], [131]–[136]
Siena	yes	sEEG	512	29	10	[117]–[123], [137]–[139]
EPILEPSIAE	no	iEEG	256,512	20-122	9	[99], [107], [140]–[146]
Freiburg	no	iEEG	256	64,128	8	[110]–[112], [147]–[151]
EpilepsyEcosystem	no	iEEG	400	16	5	[99], [108], [152]–[154]
SWEC-ETHZ	yes	iEEG	256	19-21	5	[124]–[127], [155]
Bonn	yes	sEEG, iEEG	173.61	1	4	[102], [103], [156], [157]
Melbourne	no	iEEG	5000	16	3	[100], [101], [158]
AES-NIH	no	iEEG	250,512	16	2	[135], [136]
MSSM	no	sEEG	256	22	2	[115], [159]
NeuroVista	no	iEEG	400	16	2	[160], [161]
Conegliano	no	sEEG	256	20	2	[105], [106]
Phoenix	no	sEEG	256	26	1	[162]
TUSZ	yes	sEEG	250,256	19	1	[129]
Salpêtrière	no	iEEG	/	/	1	[163]
MLSPred-Bench	yes	sEEG	250,500	23-31	1	[164]
HFH	no	iEEG	500-5000	/	1	[165]
Zenodo	yes	sEEG	512	29	1	[139]
SNUCH	no	iEEG	1600	24-78	1	[166]
VBH	no	Ear-EEG	250	5	1	[167]
Coimbra	no	sEEG	0.1-2000	/	1	[168]
XMU-FAH	no	sEEG	500	22	1	[169]
UW	no	iEEG	1000	64	1	[130]
SH-SDU	no	sEEG	500	18	1	[116]
SYSU	no	sEEG	500	22	1	[128]
UCI	yes	sEEG	173.61	128	1	[123]
Beirut	yes	sEEG	500	21	1	[123]
UCLA	no	sEEG	2000	19	1	[114]
Montefiore	no	sEEG	1024	19	1	[114]
IKCU	no	sEEG	100	18	1	[113]
FH-PKU	no	sEEG	500	19	1	[109]
CMU-XWH	no	iEEG	500	/	1	[104]
CCEMRC	no	iEEG	/	/	1	[170]
AMC	no	sEEG	200	32	1	[171]
Dresden	no	iEEG	500,1000	58-107	1	[136]
ZUH-KCL	no	iEEG	207	2	1	[172]

clinical interpretability and deployment, focusing on how models have been evaluated or implemented in real-world settings. This structured analysis aims to highlight common practices, emerging trends, and methodological gaps within the field.

A. DATASETS

An essential foundation of EEG-based epileptic seizure prediction research is the datasets used for model development and evaluation. In our review of 149 primary studies, we observed substantial heterogeneity in both the EEG recording modalities and the accessibility of the underlying datasets. Specifically, scalp electroencephalography (sEEG) was the most frequently used dataset type, appearing in 87 studies (58.4%), reflecting its popularity due to noninvasive nature and the presence of multiple publicly available repositories. Intracranial electroencephalography (iEEG) was utilized in 29 studies, while 32 studies (21.5%) combined iEEG and sEEG to leverage their complementary spatial coverage and signal characteristics. Only a single study (0.7%) employed ear-EEG [167], an emerging wearable modality that offers portable monitoring but remains underexplored in the seizure prediction literature.

With respect to dataset accessibility, 100 of the 149 studies (67.1%) leveraged open-source datasets, thereby supporting reproducibility and enabling cross-study benchmarking. In contrast, 29 studies (19.5%) relied exclusively on non-public clinical datasets, which were typically acquired in specialized epilepsy monitoring units. An additional 20 studies (13.4%) combined both open and non-open datasets, reflecting a growing effort to improve generalizability by augmenting standardized benchmark data with real-world clinical recordings. Table 2 presents an overview of the datasets used in EEG-based seizure prediction studies; individual papers may appear in multiple categories if they incorporate more than one dataset.

As shown in Table 2, several public datasets have been widely adopted in recent studies. The most frequently used is Children's Hospital Boston–Massachusetts Institute of Technology Scalp EEG Dataset (CHB-MIT), cited in 107 studies. It contains sEEG recordings from 23 pediatric patients (aged 1.5 to 19 years) acquired with 23 channels at 256 Hz. In total, 916 h of data were collected over 2 to 4 monitoring days per subject (median 4 days), stored in 664 EDF files (about 29 one-hour segments per subject). Annotations by clinicians indicated 198 clinical seizures, averaging

8.6 seizures per subject. The second most used public datasets is American Epilepsy Society Seizure Prediction Challenge Dataset (AES-Kaggle), cited in 24 studies. It provides long-term iEEG from 7 subjects—2 human patients and 5 epileptic dogs—recorded on 15 to 24 channels. In total, 1333.7 h of data were acquired, sampled at 5 kHz for humans and 400 Hz for canines. Recordings are provided as 1-h files that are further subdivided into 6 non-overlapping 10-min clips, yielding 4067 segments with 111 annotated seizures (one preictal clip per seizure and 3956 interictal clips). Siena Scalp EEG Dataset (Siena EEG), referenced in 10 studies, contains adult scalp recordings from 14 patients (ages 20 to 71) recorded at 512 Hz with 19 to 21 international 10-20 leads. Across the cohort, it includes 128 h of video-EEG (mean 9.1 h per subject), collected over 1 to 5 sessions per patient, with 47 clinically annotated seizures in total. SWEC-ETHZ Intracranial EEG Dataset (SWEC-ETHZ), used in 5 studies, comprises continuous long-term recordings from 18 drug-resistant epilepsy patients, sampled at 512 to 1024 Hz via 24 to 128 implanted electrodes. It provides 2664 h of iEEG containing 116 annotated seizures—approximately 148 h per subject—with data stored as hourly files.

In contrast, non-public datasets remain indispensable for studies requiring high-quality, long-duration, or high-channel-count iEEG recordings. The most frequently used non-public datasets is European Epilepsy Dataset (EPILEPSIAE) (nine studies), which aggregates long-term surface and iEEG from 275 patients, each monitored for at least 96 h, yielding over 30000 h of data and approximately 1800 clinically annotated seizures; recordings include up to 122 channels sampled at rates of up to 2500 Hz. Freiburg EEG Dataset (Freiburg) (eight studies) comprises depth electrode recordings from 21 drug-resistant epilepsy patients using 64 or 128 channels. In total, 509 h of data were collected over 3 to 10 days per subject (median 5 days), stored as 509 one-hour segments, and annotated for 87 clinical seizures (mean 4.1 per patient, range 1 to 11). Epilepsy Ecosystem Platform (EpilepsyEcosystem) (five studies) provides iEEG from 18 drug-resistant epilepsy patients recorded on 24 to 128 implanted electrodes at 512 to 1024 Hz; the datasets comprises 2656 h of data (median 148 h per subject), stored as one-hour segments, with 116 clinically annotated seizures (mean 6.4 per patient, range 1 to 24).

Lastly, some studies adopt an integrative strategy that leverages both public and non-public datasets to enhance model effectiveness and clinical relevance. For example, Chen et al. [109] validated their statistical method on both CHB-MIT and a non-public sEEG dataset from Peking University First Hospital (FH-PKU), achieving high accuracy with interpretability. Guo et al. [104] pretrained their model on CHB-MIT and fine-tuned it using non-public intracranial EEG recordings from Xuanwu Hospital, Capital Medical University (CMU-XWH), which improved generalization. Shafiezadeh et al. [105] trained on CHB-MIT and calibrated their model with non-public sEEG data collected at E. Medea Hospital in Conegliano, Italy, boosting performance with minimal patient-specific data. These studies demonstrate that combining CHB-MIT with institution-specific clinical recordings across modalities enhances both predictive performance and practical applicability.

Across the 149 studies reviewed, the dataset landscape is heavily skewed toward a small set of public sEEG dataset. 67.1% of papers rely exclusively on open repositories, and the pediatric CHB-MIT dataset alone constitutes 71.8% of all dataset citations. By contrast, long-duration, high-density intracranial collections such as EPILEPSIAE and Freiburg appear in fewer than one in ten studies, and only 13.4% of investigations fuse public benchmarks with center-specific clinical data. This pronounced imbalance implies that many headline performance numbers are optimized for low-channel, child-specific signal characteristics and may not generalize to adult populations, high-frequency iEEG, or real-world ambulatory recordings. Reported "state-of-the-art" results should therefore be interpreted as best-case outcomes within a narrow data regime, not as evidence of broad clinical robustness.

B. EEG SIGNAL PREPROCESSING METHODS

Preprocessing is a foundational step in EEG-based seizure prediction because it shapes feature fidelity and therefore downstream model performance. This section focuses on four core components of the preprocessing pipeline: segment division and preictal definition, denoising methods, and data balancing strategies.

1) SEGMENT DIVISION AND PREICTAL DEFINITION

Of the 149 studies, epoch length is explicitly reported in 131 studies (87.9%), with a clear preference for short segments. The most frequently adopted lengths are 5 to 10 s (50 studies, 33.6%), followed by durations shorter than 5 s (36 studies, 24.2%) and those between 20 and 30 s (35 studies, 23.5%). Longer segments exceeding 30 s are relatively rare, appearing in only 8 studies (5.4%). In addition, six studies (4.0%) employ a multi-scale design, systematically evaluating multiple segment lengths (e.g., from 5 to 60 s) within the same work to characterize the trade-off between temporal resolution and prediction accuracy. These exploratory efforts underscore the importance of segment length selection in balancing model precision and real-time applicability and indicate that future research should further refine optimal segment configurations.

Regarding the preictal prediction horizon, 125 studies (83.9%) explicitly specify a single prediction window. The most common choice is the 30 to 60 min window (91 studies, 61.1%), with a 30-min horizon alone used in 61 studies (40.9%), reflecting a prevailing compromise between clinical usefulness and predictive reliability. Shorter horizons ($\leq$10 min) are less frequent (17 studies, 11.4%), as are intermediate windows of 15 to 20 min (six studies, 4.0%); horizons longer than 60 min appear rarely (four studies, 2.7%). Nine studies (6.0%) adopt multi-scale designs, systematically evaluating performance across a range of

TABLE 3. Overview of denoising methods applied during EEG preprocessing (cutoff frequencies in Hz).

Type	Cutoff	Freq.	Ref.
Low-pass	≤ 30	3	[34], [123], [138]
	31–60	14	[37], [55], [71], [78], [79], [98], [99], [117], [126], [139], [166], [167], [169], [172]
	61–100	8	[25], [31], [39], [120], [121], [137], [165], [171]
	> 100	13	[44], [47], [49], [61], [63], [105], [106], [115], [131], [141], [145], [155], [160]
High-pass	≤ 0.5	27	[25], [31], [39], [44], [55], [61], [63], [71], [79], [98], [99], [117], [120], [121], [123], [131], [137], [139], [141], [144], [145], [155], [165]–[167], [171], [172]
	0.6–1	7	[32], [72], [78], [81], [105], [106], [169]
	1–3	2	[126], [129]
	> 3	3	[34], [47], [138]
Notch	50	22	[25], [32], [36], [37], [70], [72], [81], [105], [106], [108], [110]–[112], [119], [121], [142], [144]–[147], [150], [165]
	60	26	[36], [37], [40], [41], [43], [46], [50], [56], [60], [66], [70], [75], [80], [85], [90], [93], [108], [110]–[112], [114], [119], [122], [125], [141], [171]
	100	9	[32], [72], [105], [106], [108], [110]–[112], [150]
	120	17	[40], [41], [43], [46], [50], [56], [66], [75], [80], [85], [90], [93], [108], [110]–[112], [125]
Unspecified	/	10	[26], [33], [35], [64], [68], [82], [86], [88], [92], [132]
Others	/	7	[48], [57], [59], [62], [133], [136], [152]

TABLE 4. Overview of EEG seizure prediction data balancing strategy.

Strategy	Freq.	Ref.
Interictal undersampling	57	[24], [30], [38], [39], [43], [47], [51], [58], [60], [61], [63]–[65], [67], [70], [71], [73], [76], [78]–[84], [92], [98], [99], [104], [105], [109]–[111], [113], [115], [117], [121], [125], [126], [128], [129], [132], [133], [135], [138]–[140], [143], [146], [149], [151], [164]–[167], [169], [171]
Preictal overlapping oversampling	19	[27], [29], [34]–[36], [41], [56], [72], [85], [88], [93], [101], [103], [112], [114], [118], [150], [154], [158]
Preictal overlapping oversampling + Interictal undersampling	14	[46], [55], [66], [74], [89], [91], [94]–[96], [122], [124], [127], [134], [160]
SMOTE	4	[32], [152], [153], [163]
Preictal duplication oversampling	4	[40], [75], [142], [147]
Synthetic preictal	4	[52], [86], [97], [108]
Preictal active upsampling	3	[57], [116], [172]

windows (5 to 120 min) to characterize model sensitivity and robustness to temporal specification. These results indicate that preictal horizon selection should be adapted to the clinical context, balancing early warning lead time against prediction accuracy.

2) DENOISING METHODS

Of the 149 studies, 60 (40.3%) skip explicit denoising. Two studies—Halawa et al. [42] and Wu et al. [135]—employed advanced signal-based denoising techniques, namely discrete wavelet transform (DWT) soft-thresholding and multidimensional singular value mode decomposition (SVMD), respectively. The remaining 87 studies (58.4%) rely on classical filtering approaches, applying one or more of low-pass, high-pass, or notch filters during preprocessing prior to any feature extraction. Table 3 summarizes these approaches, because some studies employ multiple filter types, and individual papers may be counted in more than one category.

In epileptic seizure prediction research, filter cutoff frequencies reflect the trade-off between preserving informative EEG components and suppressing noise. As summarized in Table 3, the most common low-pass cutoff lies between 31 and 60 Hz (14 studies), balancing attenuation of high-frequency artifacts such as electromyography (EMG) contamination with retention of seizure-related β- and γ-band activity. Another eight studies use cutoffs in the 61 to 100 Hz range, and 13 studies employ cutoffs above 100 Hz, indicating that some deep learning approaches attempt to exploit ultra–high–frequency features. Only three studies apply a low-pass cutoff of 30 Hz or lower. High-pass filtering exhibits a consistent practice: 27 studies set the cutoff at or below 0.5 Hz to remove direct current (DC) drift and slow baseline fluctuations, while an additional seven studies employ cutoffs in the 0.6 to 1 Hz range to provide further attenuation of residual very low-frequency noise.

Notch filtering is predominantly used to suppress power-line interference, with 50 Hz and 60 Hz notches apply in 22 and 26 studies, respectively, reflecting widespread agreement on removing mains contamination. A further nine and 17 studies employ notches at 100 Hz and 120 Hz, respectively, to target higher-order harmonics and thereby attenuate residual line-related artifacts. 10 studies apply broader filter types (e.g., band-pass or band-stop) without reporting specific cutoff frequencies, and seven others rely on alternative denoising strategies, including extended training segments [48], independent component analysis (ICA) [57], Kalman filtering [59], wide-bandwidth filtering as in Whitehead's approach [62], Fourier-domain smoothing [133], manual artifact-channel rejection [136], and principal component analysis (PCA) [152].

3) DATA BALANCING STRATEGIES

EEG-based seizure prediction routinely confronts severe class imbalance, since interictal segments vastly outnumber preictal ones. To mitigate this, a variety of data-balancing strategies have been employed.

As shown in Table 4, interictal undersampling is the most prevalent data-balancing strategy (57 studies), reducing the majority class size to match preictal data and thereby mitigating classifier bias. 19 studies expand the minority class through overlapping preictal segments generated by sliding windows, while 14 combine undersampling and oversampling to both balance class proportions and preserve temporal resolution.

Some investigations explore synthetic augmentation: four studies apply synthetic minority over-sampling technique (SMOTE) and another four perform straightforward preictal duplication. A smaller subset develops bespoke reconstruction schemes, partitioning existing preictal periods into short subsegments (e.g., 1 s) and randomly reassembling them to approximate incomplete or sparse preictal data. More recent efforts (four studies) employ generative methods—such as generative adversarial networks (GANs) or temporal neural data augmentation (TNDA)——to synthesize diverse and physiologically plausible preictal examples.

These methodological pipelines are implemented in diverse software environments. Among the 149 studies, 74 (49.7%) use Python, 23 (15.4%) use MATLAB, five employ hybrid Python–MATLAB workflows, and two rely on R. Fewer than 10% of the studies included direct links to their code repositories in the main text.

In most studies, EEG recordings are divided into short epochs of 10 s or less, and a preictal prediction horizon of 30 to 60 min is the most common choice. A large fraction of the study relies solely on basic notch or band-pass filtering for signal denoising. Class imbalance is chiefly mitigated via interictal undersampling; only a minority of studies incorporate oversampling or synthetic data augmentation. While these convenient defaults simplify training and reduce methodological complexity, they frequently neglect longer-term prodromal dynamics, constrain the available contextual background, and can elevate false-alarm rates in noisy, continuous recordings.

C. MODEL STRUCTURE

This section examines the structural design of seizure prediction models in the reviewed studies, organized into four main categories: traditional machine learning models, deep learning models, studies that compare the two approaches, and hybrid models that integrate both. These categories reflect the dominant modeling strategies adopted in current research.

1) CONVENTIONAL MACHINE LEARNING FRAMEWORKS

Among the reviewed studies, 22 adopt conventional machine learning approaches. This subsection outlines the feature extraction methods and the core algorithms employed in these frameworks.

a: FEATURE EXTRACTION

Table 5 summarizes the feature extraction methods used in the 22 studies adopting conventional machine learning frameworks for seizure prediction. All of these studies rely on manually extracted features, reflecting the dominant practice in traditional machine learning approaches.

As summarized in Table 5, time- and frequency-domain features remain foundational in machine learning-based seizure prediction, appearing in 14 and 15 studies respectively, and reflecting a reliance on basic statistical descriptors and spectral analyses. Time-domain features——mean, variance, peak-to-peak, zero-crossing rate, skewness, and kurtosis—directly characterize EEG amplitude dynamics. Frequency-domain methods typically apply fast Fourier transform (FFT) to convert signals [68], estimate power spectral density (PSD) (using classical approaches and Welch's method) to quantify δ, θ, α, β, and γ band energies [81], and employ DWT for multi-scale decomposition [140]. Joint time–frequency representations—such as the wavelet transform (WT) [68], [120], [144], stationary wavelet transform (SWT) [151], and synchrosqueezing transform (SET) [102])—capture combined temporal and spectral information but, due to higher computational cost, appear in only nine studies. Nonlinear complexity metrics (sample entropy, approximate entropy, Lempel–Ziv complexity) are employed in just 4 studies, suggesting that their potential for detecting preictal chaotic dynamics remains underexplored. Connectivity features— phase-locking value (PLV) [163], phase lag index (PLI), weighted phase lag index (WPLI) [138], and graph-theoretical measures [109]—are emerging in about five studies: phase-synchrony indices are first computed between channel pairs, then converted into brain networks from which node degree, clustering coefficient, and global efficiency are extracted. Finally, second-level cross-domain fusion features generated via genetic algorithms represent an advanced trend in automated feature optimization, combining multiple primary feature sets to enhance predictive performance.

Notably, Chen et al. [109] introduced an additional processing stage in which Graphical Lasso was applied to estimate dynamic functional connectivity, producing a sparse brain network from which topological and connectivity features were subsequently derived.

b: CORE ALGORITHM

The second part of this section examines the core algorithms used in conventional machine learning pipelines. Among the 22 included studies, support vector machine (SVM) are the most frequently used, followed by LR, k-nearest neighbors (k-NN), and extreme gradient boosting (XGBoost). As shown in Table 6, the usage frequency and representative references for each algorithm are clearly presented.

TABLE 5. Overview of feature extraction methods in machine learning-based seizure prediction.

Domain	Freq.	Common Methods	Ref.
Time-domain features	14	Statistical analysis	[45], [68], [81], [99], [106], [107], [120], [132], [138], [140], [143]–[145], [167]
Frequency-domain features	15	DWT (frequency components), FFT, power spectral analysis, Welch method	[45], [68], [81], [99], [106], [107], [119], [130], [132], [138], [140], [143]–[145], [167]
Time–frequency-domain features	9	DWT, WT, SWT, SET	[68], [78], [99], [102], [107], [120], [143], [144], [151]
Nonlinear features	4	Nonlinear complexity analysis	[106], [107], [120], [167]
Spatial structure / Connectivity features	5	PLV, PLI, WPLI, graph construction & graph-theoretical metrics	[106], [109], [138], [156], [163]
Second-level features	2	Cross-domain fusion based on genetic algorithms	[144], [145]

TABLE 6. Overview of core algorithms in machine learning–based seizure prediction.

Algorithm	Freq.	Ref.
SVM	8	[45], [68], [78], [107], [132], [140], [143], [163]
LR	4	[99], [109], [144], [145]
k-NN	2	[102], [156]
XGBoost	2	[81], [106]
Others	6	[119], [120], [130], [138], [151], [167]

Among the reviewed studies, SVM emerges as the most frequently applied classifier, appearing in 8 studies; it is valued for its capacity to maximize class separation via kernel-induced mappings and its robustness in high-dimensional, small-sample regimes. LR follows as a lightweight, interpretable linear model used in four studies, both as a standalone baseline and as a component within ensemble constructions. k-NN appears in two studies; despite its simplicity and absence of an explicit training phase, it remains effective in lower-dimensional feature spaces or modest dataset sizes through straightforward distance-based voting. Finally, XGBoost is adopted in two studies to capture complex nonlinear relationships, with built-in regularization that mitigates overfitting—particularly potent when EEG features are preextracted into a compact representation. Beyond these mainstream models, six studies each introduce or adapt a distinct classification strategy. One employs a shallow multi-layer perceptron (MLP) to illustrate how slowly varying noise patterns in EEG can be misinterpreted as seizure activity, underscoring the need for rigorous data validation [130]. Another uses random forest (RF) on fused frequency-domain and phase-coupling features, achieving high accuracy through ensemble voting over multiple decision trees [119]. A third study implements boosted trees (BT) in a lightweight configuration tailored to wearable devices, balancing prediction performance with real-time constraints [167]. Kapoor et al. [120] designed a hybrid ensemble—tuned via a novel seek-optimization routine—that combines AdaBoost, RF, and single decision trees to further elevate accuracy on standard benchmarks. A low-complexity, threshold-based classifier leverages phase-locking indices for real-time feature selection, making it well suited for on-device deployment [138]. Finally, a discrete hidden Markov model coupled with Wilks' Lambda feature selection models the EEG as a sequence of latent states, capturing temporal dependencies that purely static classifiers can not [151].

Notably, although using linear models as baselines is essential for meaningful comparison, only a handful of studies have quantified how much advanced models improve upon these linear benchmarks. For example, in Qureshi et al. [45] the RBF kernel SVM reached an accuracy of 94.94% and a sensitivity of 97.43% representing increases of 3.18 and 5.20 percentage points over the linear kernel SVM respectively; In Shafiezadeh et al. [106] the XGBoost classifier achieved a mean accuracy of 78.75% under random cross validation improving performance by nearly 28 percentage points over the baseline logistic regression model; In Ramachandran et al. [132] the linear kernel SVM outperformed both logistic regression and LDA in classification accuracy on the same feature set.

2) COMPARATIVE EVALUATIONS OF MACHINE LEARNING AND DEEP LEARNING MODELS

Among the reviewed studies, nine conduct comparative evaluations between deep learning and conventional machine learning approaches. This subsection summarizes the feature extraction techniques and core algorithms used in these studies.

a: FEATURE EXTRACTION

Table 7 summarizes the feature extraction methods used in the nine studies that conduct comparisons between deep learning and conventional machine learning approaches. In all of these studies, the machine learning components rely entirely on manually extracted features. For the deep learning components, five studies employ fully automated feature extraction through the model itself, while the other four studies incorporate manually extracted features as input, followed by additional feature learning within the model [30], [76], [131], [168].

Across the nine comparative studies, spectral features proved to be the workhorse for both machine learning

TABLE 7. Overview of feature extraction methods in machine learning vs. deep learning seizure prediction studies.

Domain	Freq.	Common Methods	Ref.
Time-domain features	7	Statistical analysis	[28], [30], [74], [105], [136], [153], [168]
Frequency-domain features	8	DFT, EMD, power spectral analysis	[28], [30], [74], [105], [131], [136], [153], [168]
Time–frequency-domain features	1	WT, DWT	[168]
Nonlinear complexity features	2	Nonlinear complexity analysis	[76], [136]
Spatial structure/connectivity features	2	Functional connectivity network topology extraction	[30], [136]

and deep learning pipelines: eight of the studies rely on hand-crafted frequency-domain descriptors—discrete Fourier transform (DFT) [131], empirical mode decomposition (EMD) [74], power spectral analysis [136]—to characterize preictal versus interictal EEG. Time-domain statistics—mean, variance, skewness, kurtosis, and the like—are nearly as ubiquitous, appearing in seven studies as the foundation for conventional classifiers and sometimes as inputs to hybrid deep learning models. In contrast, joint time–frequency features are relatively rare, adopted by only one study, suggesting a trade-off between analytic richness and complexity. A smaller subset of two studies employ nonlinear complexity measures (e.g., entropy and fractal dimension) to characterize dynamical irregularities beyond what linear spectral methods capture, while two other investigations explore network-level topology by deriving functional connectivity and graph-theoretic metrics.

In addition, only one study [131] transform EEG segments into short-time Fourier transform (STFT) spectrograms before feeding them into a convolutional neural network (CNN), whereas the other five studies supply their inputs directly to deep learning models without constructing any time–frequency image representations.

b: CORE ALGORITHM

Among the nine studies that set machine-learning methods head-to-head with deep learning models, four carry out the comparison under clearly defined, often restrictive experimental scenarios. Only Gao et al. [28] employed a traditional linear SVM as their baseline, applying a genetic algorithm-based sample-weighting framework on the CHB-MIT dataset and demonstrating that the Transformer achieved the highest sensitivity and accuracy. By contrast, Shafiezadeh et al. [105] compared CNN with XGBoost classifier using both randomized cross-validation and leave-one-patient-out schemes, finding that the CNN consistently achieved slightly greater accuracy gains than XGBoost; Feizbakhsh and Omranpour [76] extracted cluster-based phase-space density features via k-means on reconstructed EEG windows, then fed these into RF and CNN, observing that the CNN performed marginally better than the RF. Müller et al. [136] contrasted a shallow MLP with CNN and examined whether false alarms from fundamentally different pipelines align temporally, thereby probing the validity of the fixed 65- to 5-min preictal assumption.

In the remaining five studies conducted under identical experimental conditions, only Sun et al. and Abdellatef et al. explicitly included simple linear classifiers as baseline models: Sun et al. [131] compared LDA and LR with CNN and recurrent neural network (RNN), finding that the CNN achieved an overall accuracy at least five percentage points higher than the linear models and Abdellatef et al. [74] included LR as a linear baseline, alongside HMS-feature–based SVM and k-NN, when comparing these machine learning methods with their residual CNN and other deep learning models, and found that HMS-SVM nonetheless outperformed the CNN. By contrast, Massoud et al. [153] evaluated SVM and random under-sampling boost alongside temporal convolutional neural network (TCNN), CNN, and long short-term memory (LSTM), reporting that SVM attained the highest accuracy in the cross-patient general model, whereas TCNN led in average accuracy following patient-specific fine-tuning; Kannan et al. [30] compared three conventional classifiers—repeated incremental pruning to produce error reduction (RIPPER), decision tree, and SVM—against a CNN and several LSTM variants; the deepest LSTM variant achieved the highest overall accuracy; Sidaoui and Sadouni [168] compared SVM with two CNN architectures, and the CNN exhibited slightly superior performance and faster inference.

Of the studies conducting head-to-head comparisons between machine learning and deep learning, only three include linear models as baselines, emphasizing that benchmarking against simple linear reference models remains uncommon.

3) HYBRID ARCHITECTURES INTEGRATING MACHINE LEARNING AND DEEP LEARNING

Among the reviewed studies, 11 employ hybrid approaches. This subsection outlines the feature extraction methods and core algorithms used in these combined frameworks.

a: FEATURE EXTRACTION

Among the 11 hybrid studies, six studies rely on end-to-end deep architectures that automatically learn features directly from their inputs, without any handcrafted preprocessing. Of these six studies, four studies specifically convert raw EEG into a time-frequency representation prior to model ingestion: Usman et al. [26], Parveen et al. [62], Toraman [67], and Esmaeilpour et al. [70]—all first convert EEG windows into 2D spectrograms via STFT or Fourier transform (FT), then feed those images into CNN-based architectures for automatic feature learning and seizure prediction. The remaining two deep models in this group bypass any time-frequency imaging and consume either raw EEG snippets or prevectorized feature arrays directly.

In a distinct ensemble strategy, Reuben et al. [161] aggregated the outputs of multiple crowd-sourced machine learning and deep learning models instead of relying on raw EEG signals. Each base model produced a time series of "preictal probability" estimates; these series are concatenated into a single composite probability vector that served as input to a downstream MLP. The final MLP was trained to learn optimal combination weights over the constituent model outputs, yielding a fused forecast that improves overall seizure prediction accuracy.

The remaining four studies employ a hybrid pipeline in which EEG segments are first represented by a suite of manually engineered features; these feature vectors are then input to deep neural networks to perform a second-stage, automated representation learning. This two-tiered approach enables the models to refine the initial handcrafted descriptors, learning higher-order abstractions and compensating for their limitations. In Usman et al. [86], each 30-s EEG window was first decomposed using empirical mode decomposition, after which a rich set of features was extracted: time-domain statistics such as mean, variance, skewness and kurtosis; spectral descriptors including centroid, variance, skewness and kurtosis computed on the intrinsic mode functions; entropy-based metrics; autoregressive prediction errors; Hjorth parameters; and various energy-related measures. Saadoon et al. [37] extracted five band-limited amplitude profiles (δ through γ) via normalized STFT and inter-channel Pearson correlations from each 10-s epoch. Hosseini et al. [165] employed Daubechies-4 wavelet decomposition to extract detail coefficients at levels 4 to 6 and computed multidimensional handcrafted features—including time-domain statistics, frequency descriptors, entropy and fractal dimensions, Hjorth parameters, and autoregressive prediction errors—then rearranged these feature vectors into "feature maps" which, together with three-channel EEG images generated from $\theta/\alpha/\beta$ band power topographies, were fed into a convolutional neural network for deep representation learning. Finally, Zargar et al. [146] computed 22 univariate measures—including time-domain moments, spectral moments, entropy, correlation dimension, and Lyapunov exponents—for each 5-s epoch; the resulting 22-dimensional vectors were reshaped into 2D "feature images" and passed through ImageNet-pretrained CNN backbones (e.g., Xception, MobileNet-V2, EfficientNet-B0) to learn higher-order representations that were then classified either by a fully connected head or by SVM.

b: CORE ALGORITHM

In this section, 11 studies are presented that combine machine learning and deep learning techniques. Eight of these employ a canonical two-stage framework—first using a deep learning model for automatic feature extraction and then a traditional machine learning classifier for prediction.

Among these eight studies, six studies propose a single hybrid machine learning and deep learning algorithm: five utilize SVM as the classifier [26], [37], [62], [165], and one uses Bayesian linear discriminant analysis (BLDA) [148]; on the feature-extraction side, four employ CNN [26], [148], [154], [165], one employs pretrained EfficientNet-B0 network [37], and one adopts dense layered network model (DLNM) [62]. The remaining two studies construct multiple hybrid deep learning and machine learning pipelines: Toraman [67] proposed three hybrid frameworks that combine pretrained 2D CNN with SVM classifiers, where each model first extracts deep spectral features via the CNN and then feeds the resulting feature vectors into an SVM to distinguish preictal from interictal states; Zargar et al. [146] proposed nine concise hybrid pipelines combining pretrained CNN for feature embedding with traditional machine learning classifiers to discriminate preictal from interictal segments. In contrast to the canonical two-stage hybrids described above, the remaining three studies employ more intricate ensemble strategies. Usman et al. [86] outputed meta-learn from SVM, CNN, and LSTM base learners using model-agnostic meta-learning (MAML); Reuben et al. [161] ensembled 8 crowd-sourced seizure prediction pipelines via a shallow, lightweight MLP; and Esmaeilpour et al. [70] applied parallel classifiers—a fully connected network, RF, and SVM—on embeddings extracted from a CNN and fuse their predictions through majority voting.

Among these hybrid approaches, only Parveen et al. [62] proposed a novel method combining DLNM with linear SVM, using pure linear SVM as the baseline and increasing average accuracy on the CHB-MIT dataset from 95.6% to 96.0%, an improvement of 0.4 percentage points.

4) DEEP LEARNING ARCHITECTURES

Among the reviewed studies, 107 adopt deep learning architectures. This subsection outlines the feature extraction methods and the core algorithms employed in these frameworks.

a: FEATURE EXTRACTION

Of the 107 studies leveraging deep learning architectures, a majority—75 studies—eschew any manual feature engineering, instead relying wholly on their models' internal representation learning. Within this subset of 75 end-to-end approaches, 31 studies operate directly on raw EEG signals or minimally preprocessed time series. The remaining 44 studies incorporate an explicit format conversion step to generate two-dimensional spectrograms or scalograms that then serve as the network's input. Table 8 provides a detailed breakdown of these distinctions.

TABLE 8. Overview of format conversion in end-to-end deep learning studies.

Method	Freq.	Ref.
STFT	21	[36], [40], [41], [44], [46], [56], [60], [66], [71], [88], [90], [93], [108], [112], [124], [125], [133]–[135], [166]
CWT	4	[84], [100], [115], [155]
S-transform	3	[79], [116], [117]
DWT	2	[34], [53]
FFT	2	[33], [172]
MFCC	2	[80], [82]
Pearson correlation	2	[54], [92]
Others	8	[57], [97], [103], [113], [122], [139], [142], [159]

Among the reviewed studies, STFT—which partitions EEG recordings into overlapping windows and computes a localized spectrum for each—is by far the most common time-frequency preprocessing step, appearing in 21 papers. Continuous wavelet transform (CWT), used in four studies, offers a multiscale time-frequency decomposition by correlating the signal with scaled and shifted mother wavelets. The S-transform, reported in three studies, combines elements of STFT and CWT by applying a frequency-dependent Gaussian window, thereby achieving improved spectral resolution. DWT, cited in two studies, hierarchically decomposes the signal into sub-bands aligned with canonical EEG rhythms. FFT—an efficient algorithm for obtaining the global spectrum—and mel-frequency cepstral coefficient (MFCC)—which capture the spectral envelope on a perceptually motivated scale—are each applied in two studies. Finally, two studies construct Pearson-correlation-based brain-network graphs from pairwise channel correlations and derived connectivity features for classification.

In addition to these dominant time–frequency transforms, eight studies explore more specialized conversion techniques. One study employed global field power (GFP) to condense multichannel EEG into a single time series reflecting overall signal variance [139]. Another used ICA to isolate and remove artifacts while preserving neural components [57]. Markov transition field (MTF) was applied in one study to encode temporal transitions into image-like representations [103]. The multivariate weighted fuzzy granular recurrence plot method was used to construct graphs that capture frequency-specific connectivity patterns. [97]. Another study employed PLV analysis to quantify phase synchrony between electrode pairs, using the resulting measures as input features [122]. Power spectral density (PSD) mapping was used to capture the distribution of signal power across frequencies [142]. The synchrosqueezing transform (SST) provided sharpened time–frequency localization by reassigning spectral energy [113]. Finally, Li et al. [159] proposed a multi-metric similarity-weighted sparse electrode graph that combines several connectivity measures into a single sparse network representation, emphasizing the most salient interactions.

The remaining 32 studies adopt a two-stage feature extraction pipeline: handcrafted EEG features are computed first, and these preextracted representations are then supplied to deep learning architectures for secondary data-driven representation learning. Table 9 presents an overview of the feature extraction methods.

Among the 32 studies that employ handcrafted feature extraction before deep learning, frequency-domain features are the most prevalent, used in 25 studies. These features are typically derived through methods such as FFT [61], [126], [141], [162], power spectral analysis [59], [69], [150], [165], MFCC [29], [118], or wavelet packet decomposition [31], [32], highlighting the rhythmic structure of EEG signals and aligning well with convolutional architectures. Time-domain features appear in 14 studies, relying on basic statistical descriptors like mean, standard deviation, and skewness to capture signal amplitude characteristics. Time–frequency domain features are adopted in 10 studies, where transformations like STFT [43], [73], [85], DWT [24], [141], [152], [165], or CWT [65], [114] provide a joint temporal and spectral representation of EEG; these are especially useful for capturing nonstationary behaviors associated with preictal transitions. Nonlinear features, reflecting the complex and chaotic properties of EEG signals, are utilized in nine studies, often based on sample entropy [73], [110] or fractal analysis [39], [126], [141]. Spatial and connectivity-based features—those that represent inter-channel dependencies via functional connectivity metrics or graph-theoretical constructs—are relatively uncommon. References [24], [59], and [98], reported in only six studies. Finally, dynamic features—designed to reflect the evolution of signal properties over time—are used in just three studies. In Borhade et al.'s ResNeXt–LeNet work [59], EEG frames undergo STFT and spectral flux was computed to capture dynamic changes; in Ravindranath et al.'s MMA study [162], a multivariate channel graph was Gaussian-smoothed across scales to extract random matrix theory features; and in Assali et al.'s CNN classification [73], a Kalman-filtered multivariate autoregressive model (MVAR) model applied to 2-s windows produced an eigenvalue-based stability index that captured temporal dynamics.

Distinct from the preceding extraction methods, eight of the 32 studies augment their pipelines with additional format transformations to enrich the learned representations. Two of these first construct specialized graph structures—one by using maximum cross-correlation to generate a channel-wise brain graph [24] and the other by leveraging metadata to build a multimodal association graph [162]—and subsequently extract graph embedding features from those structures. Five others manually extract feature sets and encode them as graph-based inputs, such as spatial and functional edge-weight graphs between electrodes [32], Pearson correlation–based functional connectivity graphs [65], [150], global phase-amplitude coupling (GPAC) maps [114], and

TABLE 9. Overview of feature extraction methods in deep learning based seizure prediction.

Domain	Freq.	Common Methods	Ref.
Time-domain features	14	Statistical analysis	[24], [32], [49], [59], [64], [65], [72], [94], [123], [126], [137], [141], [152], [170]
Frequency-domain features	25	FFT, MFCC, power spectral analysis, WPT (frequency part), Welch's method	[24], [29], [31], [32], [49], [59], [61], [63]–[65], [69], [85], [94], [98], [118], [123], [126], [137], [141], [147], [150], [152], [160], [162], [170]
Time–frequency-domain features	10	DWT, CWT, WT, STFT	[24], [27], [43], [59], [65], [73], [85], [114], [141], [152]
Nonlinear features	9	Nonlinear complexity analysis	[32], [39], [64], [73], [110], [123], [126], [137], [141]
Spatial structure / connectivity features	6	Construction and graph-theoretical metrics	[24], [32], [59], [98], [141], [147]
Dynamic features	3	Multiscale dynamic temporal extraction	[59], [73], [162]

power spectral density energy diagrams [69]. The last study employ a dual-stream configuration, supplying the model with both an STFT spectrogram and manually computed feature values, thereby enriching its spatiotemporal and spectral representations [72].

b: CORE ALGORITHM

Of the 107 studies applying deep learning to seizure prediction, only Georgis–Yap et al. conducted a direct head-to-head comparison of multiple architectures, evaluating supervised CNN, CNN–LSTM hybrids, and TCN alongside their unsupervised autoencoder counterparts (CNN-AE, CNN–LSTM-AE, and TCN-AE) [124]. The other 106 studies each examine a single primary model; these algorithms are listed in Table 10. Collectively, the table provides a compact yet comprehensive overview of the purely deep learning based approaches reported in the literature.

In the single-architecture category, 76 studies each implement one core deep learning backbone. Among these, 44 CNN-based studies feature architectures ranging from lightweight 1-D [42], [84], [101], [149], to deep multi-branch 2-D [25], [29], [36], [38], [87], [100], [129], [159] and 3D–2D hybrids [88]; whether using dilated kernels, channel attention, or multi-view inputs, each relies on convolutional filters to capture spatial–spectral signatures of preictal EEG. 14 studies utilize RNN-based designs to model the long-range temporal dynamics of preictal EEG sequences; examples include two-layer LSTM [24], [61], bidirectional LSTM [31], [157], deep gated recurrent unit (GRU)–LSTM cascades [49], stacked LSTM variants [123], [158], and LSTM enhanced with metadata-supported multivariate attention [162]. Seven studies adopt Transformer-based encoders, substituting self-attention for recurrence to model global context across extensive windows.

Five studies investigate GNN–based approaches, including graph convolutional network (GCN) [32], [57], [118], joint graph structure and representation learning network [150], and graph attention network (GAT) [122]; these methods model electrodes as graph nodes and leverage message passing to infer dynamic connectivity. Four studies employ feedforward neural network (FNN)-based architectures, from deep MLP [89], [152] and deep neural network (DNN) [142] to fully connected feedforward networks [170].

Finally, one TCN-based study uses an anchoring temporal convolutional network (ATCN) with dilated causal convolutions for very long receptive fields, and one variational autoencoder (VAE)-based study employs a residual convolutional variational autoencoder that flags preictal windows as reconstruction outliers in the learned latent space.

In the hybrid-architecture group, 30 studies fuse complementary modules to harness multiple feature domains. The most common configuration—reported in 16 studies—stacks a CNN front end with an RNN back end: lightweight 1-D/2-D CNN [27], [39], [72], [98], [171] or residual variants such as ResNet-50 [44] and deep residual shrinkage network (DRSN) [50] distill spatial-spectral patterns, which are then temporally aggregated by LSTM/bidirectional long short-term memory (BiLSTM) [27], [39], [44], [52], [53], [64], [92], [169] or, less often, GRU units [35], [50]. Eight studies fuse CNN with Transformer blocks, typically using a lightweight 1-D or 2-D CNN front end [43], [55], [103] followed by a standard self-attention encoder [43], [55], [91] that refines the convolutional feature maps to capture long-range dependencies. Less common hybrids include two CNN-GNN pipelines. Dynamic functional connectivity neural network [54] is a dual-branch architecture with separate GCN and CNN streams, while spatio-temporal-spectral network [104] integrates a spatio dynamic graph convolution network branch with triple-attention and multi-scale CNN layers, thereby coupling spectral filtering with graph-based reasoning.

Four additional hybrids appear only once each, underscoring the field's exploratory breadth: a CNN-FNN network [111] augments convolutional embeddings with a Fourier neural network branch to enrich spectral cues; a restricted Boltzmann machine (RBM)-Transformer model [33] couples contractive slab-and-spike generative layers with self-attention to capture higher-order dependencies; a RNN-Transformer cascade [125] stacks a LSTM encoder beneath a Transformer, uniting sequence memory with global context; and a GNN-RNN pipeline [65] combines spatial GCN with temporal LSTM reasoning to jointly model electrode connectivity and signal evolution.

TABLE 10. Overview of core algorithms in deep learning based seizure prediction.

Algorithm	Freq.	Ref.
CNN-based	44	[25], [29], [36], [38], [42], [46], [48], [51], [56], [58], [59], [63], [69], [71], [73], [77], [83], [84], [87], [88], [90], [94]–[97], [100], [101], [108], [112], [113], [115], [121], [128], [129], [133], [137], [139], [141], [147], [149], [155], [159], [160], [166]
CNN + RNN-based	16	[27], [35], [39], [44], [50], [52], [53], [64], [66], [72], [92], [98], [114], [127], [169], [171]
RNN-based	14	[24], [31], [47], [49], [60], [61], [79], [117], [123], [126], [157], [158], [162], [172]
CNN + Transformer-based	8	[40], [43], [55], [80], [82], [91], [103], [134]
Transformer-based	7	[34], [41], [85], [93], [116], [135], [164]
GNN-based	5	[32], [57], [118], [122], [150]
FNN-based	4	[89], [142], [152], [170]
CNN + GNN-based	2	[54], [104]
TCN-based	1	[110]
VAE-based	1	[75]
CNN + FNN-based	1	[111]
RBM + Transformer-based	1	[33]
RNN + Transformer-based	1	[125]
GNN + RNN-based	1	[65]

Among the 107 studies that applied deep learning to seizure prediction, only 10 explicitly benchmark their models against simple linear baselines. Qiao et al. [33] proposed Attention-CssCDBN, raising sensitivity on the CHB–MIT dataset from 81.0% to 98.5% compared with LDA; Pandey et al. [123] developed a stacked LSTM enhanced by hybrid feature augmentation and FB-SARO optimisation, improving accuracy by 1.40% to 4.94% over LR across four public datasets; Hassoon et al. [141] introduced the HSA-CAE-CNN, achieving roughly a 9.9-percentage-point performance gain relative to the minimum-distance classifier (MDC); Tsiouris et al. [24] employed a two-layer LSTM, boosting average sensitivity and specificity by 7.2%–16.4% compared with a linear SVM over 15 to 120 min prediction horizons. Collectively, these comparisons underscore the substantial advantage of deep models in capturing the non-linear preseizure dynamics embedded in EEG signals.

5) DIMENSIONALITY REDUCTION TECHNIQUES

Of the 149 studies, 49 (32.9%) incorporate implicit dimensionality reduction directly within model architectures. These include autoencoders that impose compression via a low-dimensional bottleneck layer; convolutional neural networks that reduce representation size through spatial downsampling and 1×1 convolutional filters; and attention mechanisms that adaptively weight and aggregate the most informative feature dimensions during training.

Eight studies (5.4%) apply explicit dimensionality reduction or channel selection techniques. PCA [90], [148], [152], [167] identifies orthogonal directions that capture the greatest variance, ranks components by explained variance, and retains the minimal subset whose cumulative variance exceeds a predefined threshold, thereby denoising and reducing dimensionality. Minimum redundancy maximum relevance (mRMR) feature selection [39] iteratively chooses features that maximize mutual information with the target while minimizing redundancy among selected features, yielding a compact and informative subset. Wilks' Lambda–based selection [151] evaluates multivariate discrimination via the ratio of within-group to total variance and retains features that minimize this statistic to enhance class separability. Other studies performed channel selection using statistical or learned criteria—such as entropy (signal unpredictability), variance (signal power), kurtosis and skewness (non-Gaussian characteristics), or attention-derived weights from trained models——to identify the most relevant EEG channels for downstream analysis [35], [84]. Furthermore, two studies combine both explicit and implicit approaches: these works first applied PCA to extract the leading principal components and then embedded additional compression within the model architecture via low-dimensional bottleneck layers or 1×1 convolutional filters to further refine and compact EEG feature representations [141], [162].

Seizure prediction modelling remains overwhelmingly convolution-centric: fully 71.8% of studies rely on a pure deep learning backbone, with CNN variants alone accounting for 48.3%. Hybrid stacks occasionally append LSTM or Transformer layers to extend temporal context, yet architectures driven entirely by attention mechanisms or graphs appear only as proofs of concept. Because most pipelines convert EEG into short-time spectrograms, networks are tuned to local time–frequency textures rather than large-scale network dynamics; features such as phase synchrony or cross-channel graph structure are seldom learned directly. Dimensionality reduction is likewise predominantly implicit, achieved through architectural operations (e.g., pooling and strides) rather than explicit feature selection, which further narrows the representational scope. Compounding this methodological narrowness, only a small subset of studies benchmark their models against linear baselines, making it difficult to quantify the true incremental value of deep architectures and risking overestimation of "state-of-the-art" gains. The field thus exhibits both technical maturity—strong, repeatable CNN baselines—and methodological conservatism that

TABLE 11. Overview of SOP and SPH settings.

SOP	Freq.	Ref.
30 min	37	[36], [40], [41], [46], [50], [51], [56], [58], [60], [63], [66], [70], [79], [80], [83], [85], [87], [88], [90], [93], [95], [96], [101], [108], [112], [116], [117], [119], [122], [123], [126], [128], [134], [147], [149], [160], [169]
20 min	3	[64], [135], [143]
40 min	2	[27], [145]
50 min	2	[47], [148]
60 min	2	[102], [172]
1, 5, 10, 15, 25, 29, 120 min	1 each	[129] (1 min), [164] (5 min), [115] (10 min), [39] (15 min), [34] (25 min), [87] (29 min), [110] (120 min)
Individualised value (grid-search)	1	[99]
SPH	**Freq.**	**Ref.**
5 min	30	[34]–[36], [39], [50], [52], [56], [60], [63], [70], [79], [80], [83], [88], [90], [93], [101], [108], [112], [117], [119], [122], [126], [128], [134], [143], [147], [149], [169], [172]
10 min	9	[27], [64], [99], [102], [108], [110], [115], [145], [150]
1 min	7	[51], [87], [89], [95], [96], [160], [164]
3 min	6	[40], [41], [46], [66], [85], [116]
30 min	3	[58], [123], [135]
10 s, 0 min, 14 min	1 each	[47] (0 min), [129] (14 min), [148] (10 s)

may hamper progress toward robust, patient-independent generalisation.

D. EXPERIMENTAL VALIDATION SETUP

This section covers three core elements of our experimental validation: the definitions of seizure occurrence period (SOP) and seizure prediction horizon (SPH); the key performance metrics for assessing accuracy and timeliness; and the data partitioning schemes for training, validation, and testing. Rigorous validation is vital to prove our method's reliability and generalizability across varied EEG datasets.

1) SOP AND SPH

Of the 149 studies, 54 explicitly define and apply a SOP, whereas 58 clearly specify and implement a SPH, as summarized in Table 11.

Among the studies that explicitly report their timing configurations, two consistent patterns stand out. First, for the SOP, a 30 min window is overwhelmingly favored—37 studies adopt this setting—while alternative durations of 20, 40, 50, and 60 min appear only sporadically. Second, the SPH most commonly centers on 5 min, cited by 30 studies; the next-most-frequent choice is 10 min, reported by nine studies, whereas ultra-short windows of 1 to 3 min are rarely used. Longer or atypical settings (e.g., 10 s, 14 min, or a 30 min SPH) are exceptionally uncommon. Collectively, these findings reveal a strong convergence on clinically pragmatic defaults—namely, a 30 min SOP paired with a 5 min SPH.

2) EVALUATION INDICATORS

In this section, we analyze the evaluation metrics most frequently reported in the surveyed study. Figure 4 summarizes these metrics and depicts their prevalence across the reviewed studies.

These evaluation indicators offer deep insights into the performance of the implemented models and are widely used to assess their effectiveness. Sensitivity (SEN, see Eq. (1)) is the most frequently reported metric, appearing in 133 studies, and it intuitively reflects the probability that the predictor will catch an impending seizure: the higher the value, the fewer attacks slip through and the safer the patient. Recent experimental work generally regards a SEN of $\approx$ 0.80–0.90 as clinically acceptable. At the same time, markedly lower scores suggest a higher risk of missed events and limited practical usefulness [31], [38], [45], [64], [86], [108], [135]. The next most common metric is accuracy (ACC, see Eq. (2)), reported in 89 studies. It captures the overall share of epochs that the predictor labels correctly—whether seizure or non-seizure—so higher values imply fewer total misclassifications. On widely used public benchmarks, ACC values around 0.85 and above are typical, yet this figure alone is not enough to demonstrate practical utility unless it is supported by strong sensitivity and acceptably low false-alarm rates [42], [65], [73], [98], [125], [165]. Specificity (SPE, see Eq. (3)) ranks third with 75 studies. It expresses the probability that the predictor correctly dismisses non-seizure (interictal) segments, so higher values translate directly into fewer false alarms and less patient or caregiver disruption. In practice, seizure-prediction systems often aim for SPE above 0.90 to keep the hourly false-alarm rate within an acceptable clinical burden; if SPE drops much below that level, even a highly sensitive model can become impractical because of the volume of unnecessary warnings [54], [72], [129]. The false prediction rate per hour (FPR/h, see Eq. (4)) equals the number of false positives divided by the total monitoring hours and is noted in 69 studies. It measures how many spurious alerts the system generates each hour; lower figures reduce alarm fatigue for patients and staff. A practical benchmark is $\leq$ 0.1 alarms per hour—roughly one false alert every 10 h—to keep the burden acceptable

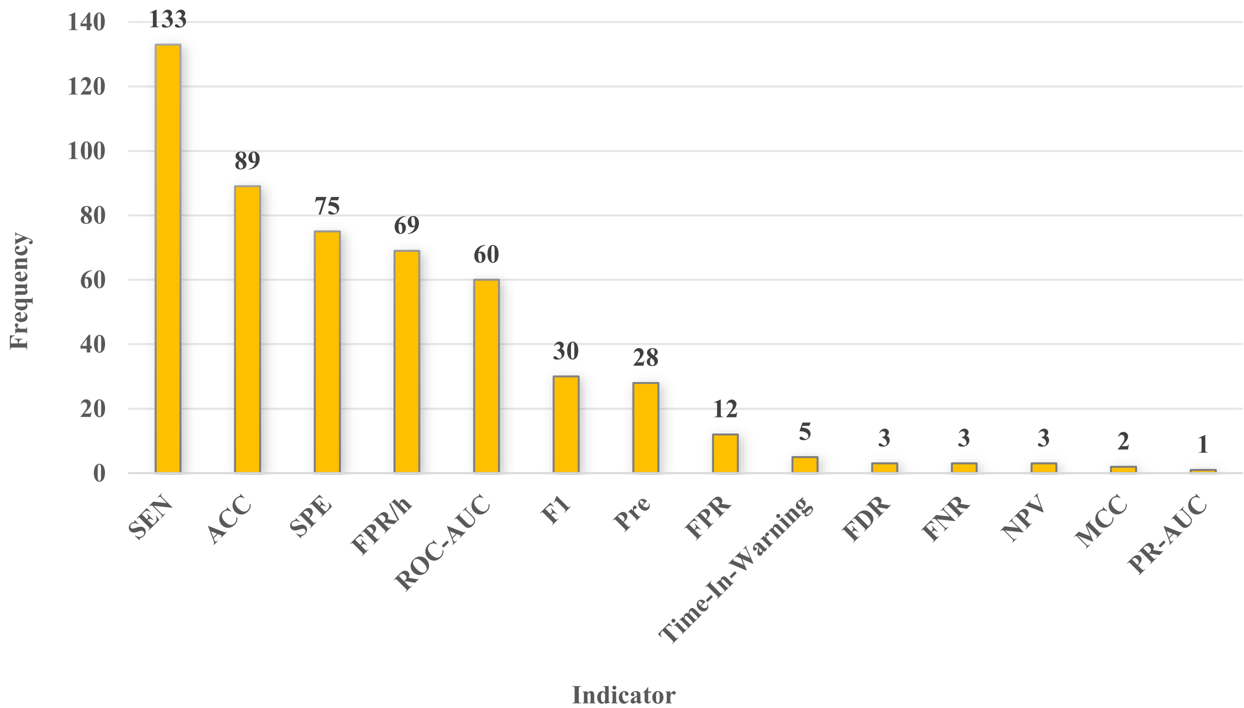

FIGURE 4. Distribution of the included studies by reported evaluation metric. SEN—sensitivity (recall or true positive rate), ACC—accuracy, SPE—specificity (true negative rate), FPR/h—false prediction rate per hour, ROC-AUC—area under the receiver operating characteristic curve, F1-score—harmonic mean of precision and recall, Pre—precision, FPR—false positive rate, Time-in-Warning—proportion of monitoring time spent in warning state, FDR—false discovery rate, FNR—false negative rate, NPV—negative predictive value, MCC—Matthews correlation coefficient, PR-AUC—area under the precision–recall curve.

in continuous monitoring [24], [54]. Area under the receiver operating characteristic curve (ROC-AUC) captures how well the model separates seizure from non-seizure cases independent of any single threshold: 0.5 indicates no better than chance, 1.0 reflects perfect discrimination. Values $\geq$ 0.90 signal strong overall performance, but ROC-AUC alone may obscure precision issues on highly imbalanced data [91], [115], [142].

Additionally, the F1-score (30 studies [98], see Eq. (5)) is the harmonic mean of precision and sensitivity. F1-score balances "how many seizures are caught" with "how many alerts are correct". It is beneficial on imbalanced datasets: values $\geq$ 0.80 suggest that the model is not favouring one class at the expense of the other; Precision (PRE, 28 studies [35], see Eq. (6)) tells how trustworthy each alarm is—the higher it is, the fewer false alerts clinicians must dismiss. Practical systems often aim for PRE $\geq$ 0.70 to keep user confidence high; False positive rate (FPR, 12 studies [44], see Eq. (7)) is the probability that a nonseizure segment is erroneously classified as preictal; lower values are preferable; Time-in-Warning is the proportion of total monitoring time that the system remains in an alert state (five studies [161], see Eq. (8)); it directly quantifies user burden. Maintaining Time-in-Warning below 20–30% is generally recommended so that patients do not live in a near-constant state of anticipatory stress.

Less frequently reported yet still important is the false discovery rate (FDR, three studies [109], see Eq. (9)), which shows what proportion of issued alerts are false. In practice, it is plotted on a maintenance dashboard; a rising trend signals that postprocessing rules or thresholds may need retuning to keep user trust high. False negative rate (FNR, three studies [165], see Eq. (10)) quantifies missed seizures——the complement of sensitivity. Deployment logs track it continuously; any sustained increase prompts checks for sensor noise, channel drop-outs, or model drift that could jeopardise patient safety. Negative predictive value (NPV, three studies [170], see Eq. (11)) reflects how reliable a "no-seizure" state is. A stable, high NPV reassures users that daily activities can continue during "safe" periods, whereas a dip typically triggers signal-quality diagnostics or recalibration, and Matthews correlation coefficien (MCC, two studies [98], see Eq. (12)) combines all four confusion-matrix terms into a single value ranging from –1 (inverse prediction) through 0 (random) to 1 (perfect). Because it remains informative even when class proportions vary, MCC is well-suited for comparing models across datasets and tracking performance over time. Area under the precision–recall curve (PR-AUC) is mentioned in only one study [124]. The area under the precision–recall curve summarises the trade-off between catching seizures (recall) and avoiding false alarms (precision) at every threshold. It is routinely reported alongside ROC-AUC during pre-deployment validation to demonstrate robustness on the highly imbalanced data typical of long-term EEG monitoring.

$$\mathrm{SEN} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{1}$$

$$\mathrm{ACC} = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN}} \tag{2}$$

$$\mathrm{SPE} = \frac{\mathrm{TN}}{\mathrm{TN}+\mathrm{FP}} \tag{3}$$

$$\mathrm{FPR}/h = \frac{\mathrm{FP}}{\text{Total monitoring hours}} \tag{4}$$

$$\mathrm{F1} = \frac{2\,\mathrm{TP}}{2\,\mathrm{TP}+\mathrm{FP}+\mathrm{FN}} \tag{5}$$

$$\mathrm{Pre} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \tag{6}$$

$$\mathrm{FPR} = \frac{\mathrm{FP}}{\mathrm{FP}+\mathrm{TN}} \tag{7}$$

$$\mathrm{Time}-\mathrm{in}-\mathrm{warning} = \frac{T_{\mathrm{warning}}}{T_{\mathrm{total}}} \tag{8}$$

$$\mathrm{FDR} = \frac{\mathrm{FP}}{\mathrm{FP}+\mathrm{TP}} \tag{9}$$

$$\mathrm{FNR} = \frac{\mathrm{FN}}{\mathrm{FN}+\mathrm{TP}} \tag{10}$$

$$\mathrm{NPV} = \frac{\mathrm{TN}}{\mathrm{TN}+\mathrm{FN}} \tag{11}$$

$$\mathrm{MCC} = \frac{\mathrm{TP}\times\mathrm{TN}\;-\;\mathrm{FP}\times\mathrm{FN}}{\sqrt{(\mathrm{TP}+\mathrm{FP})(\mathrm{TP}+\mathrm{FN})(\mathrm{TN}+\mathrm{FP})(\mathrm{TN}+\mathrm{FN})}} \tag{12}$$

where TP, TN, FP, and FN denote the counts of true positives, true negatives, false positives, and false negatives, respectively; T_{warning} is the total duration during which the system is in an alarm state, and T_{total} is the overall monitoring time.

3) DATA VALIDATION STRATEGY

Effective data partitioning is indispensable for unbiased performance evaluation and for assessing model generalization in seizure prediction research. In this section, we summarize and compare the training, validation, and test splits employed in the reviewed studies, including hold-out, k-fold cross-validation, leave-one-seizure-out, and leave-one-patient-out schemes. Table 12 presents a statistical overview of these validation strategies.

In the surveyed studies, the hold-out scheme is the most prevalent data-partitioning strategy, appearing in 57 studies (38.2%). It divides the data into fixed proportions—commonly 70% to 80% for training and the remainder for testing—without subsequent reshuffling. This single, one-time split is simple to implement and is widely used in both traditional machine learning and deep learning studies.

Leave-one-seizure-out (LOSO) is employed in 44 studies (29.5%). Under this protocol, each seizure is held out in turn as the test set, with the remaining seizures and interictal segments forming the training data. Repeating this procedure for every seizure permits evaluation of the model's ability to predict previously unseen seizure events from the same patient.

K-fold cross-validation appears in 32 studies (21.5%). The data are partitioned into K approximately equal folds, and the model is trained and evaluated K times, each time using a different fold for testing and the others for training. This scheme ensures that every sample contributes to both training and validation, yielding a more balanced and comprehensive assessment.

Leave-one-patient-out (LOPO) is used in nine studies (6.0%). In each iteration, all data from a single patient are withheld for testing while the model is trained on data from the remaining patients. LOPO directly measures cross-subject generalization and is especially important for patient-independent seizure prediction scenarios.

Another seven studies employ alternative validation schemes grouped under "Others", highlighting methodological heterogeneity beyond the mainstream. Two adopt sample-level leave-one-out (LOO), which maximizes data usage but can produce optimistic performance estimates [36], [132]. An intra-patient, instance-level, multi-round cross-validation scheme balances seizure coverage at the expense of assessing inter-patient generalization [138]. Leave-one-day-out (LODO) is used to account for day-to-day EEG variability [163], while sequential time-window analysis characterizes temporal performance trends [107]. Reuse of standardized splits from MLSPred-Bench enhances reproducibility [164], and a critical appraisal of partitioning schemes reveals common validation pitfalls [130]. Collectively, these approaches reflect efforts to mitigate methodological weaknesses and improve clinical relevance.

Validation strategies typically prioritize privilege convenience over clinical realism. A single 30 min SOP and 5 min SPH dominate timing choices, and SEN/ACC headline most scorecards, while practical metrics like false alarms per hour are reported in fewer than half of the studies. Hold-out or leave-one-seizure-out splits cover about 67.8% of studies; rigorous cross-patient leave-one-patient-out validation is used sparingly. Consequently, many impressive AUCs represent intra-subject performance on curated data rather than proof that a model will hold up across new individuals, devices, or 24-h home monitoring—an essential step for clinical adoption.

E. POSTPROCESSING DECISION STRATEGIES

After model inference, postprocessing decision strategies transform raw outputs into reliable seizure alarms. By imposing temporal consistency and filtering out isolated spurious signals, they enhance robustness and reduce the false alarm rate.

Among the 149 studies, 58 incorporate explicit postprocessing mechanisms, underscoring their essential role in turning raw model outputs into actionable seizure warnings. These mechanisms cluster into four principal categories—voting, thresholding, temporal smoothing, and advanced fusion strategies. Table 13 summarizes the specific techniques; because many studies adopt more than one approach, the method frequencies reported exceed the study count and therefore do not correspond directly to the number of studies.

TABLE 12. Overview of validation strategies.

Strategy	Freq.	Ref.
Hold-out	57	[26], [27], [30], [34], [35], [37], [41], [42], [44], [45], [49], [62], [63], [69], [71], [73], [74], [78], [79], [84], [85], [94], [99], [101], [103], [108], [109], [114], [122], [126], [128], [131], [133], [136], [137], [139], [143]–[146], [148], [151]–[155], [157], [158], [160]–[162], [165]–[168], [170], [172]
Leave-one-seizure-out	44	[28], [31], [32], [38], [40], [46], [47], [50]–[52], [54]–[57], [60], [64], [66], [70], [80], [82], [83], [87]–[93], [95], [96], [98], [110]–[112], [117], [119], [124], [125], [134], [135], [140], [141], [147], [149]
K-fold cross-validation	32	[24], [25], [29], [33], [39], [43], [48], [53], [58], [59], [61], [65], [67], [72], [75]–[77], [86], [102], [104], [115], [116], [118], [120], [123], [127], [129], [142], [150], [156], [159], [169]
Leave-one-patient-out	9	[68], [81], [97], [100], [105], [106], [113], [121], [171]
Others	7	[36], [107], [130], [132], [138], [163], [164]

TABLE 13. Overview of postprocessing decision strategies.

Category	Method	Freq.	Ref.
Voting mechanisms	k-of-n	15	[28], [40], [41], [46], [50], [56], [85], [90], [93], [112], [116], [125], [134], [135], [149]
	segment proportion	8	[47], [48], [64], [83], [114], [132], [137], [169]
	consecutive segments	6	[31], [45], [80], [89], [126], [148]
	segment maximum value	1	[172]
	channel	1	[129]
Thresholding mechanisms	fixed	21	[31], [45], [51], [57], [87], [89], [95], [96], [99], [104], [115], [140], [144], [145], [147], [148], [151], [157], [160], [163], [164]
	adaptive	1	[162]
Timing smoothing	Moving average filtering	21	[31], [45], [51], [57], [87], [89], [95], [96], [99], [104], [112], [115], [126], [140], [143]–[145], [147], [148], [160], [163]
	Refractory period	10	[28], [38], [51], [87], [93], [96], [114], [140], [145], [160]
	Sliding window	5	[64], [110], [138], [169], [172]
Adaptive ensemble mechanism	MAML, multi-classifier voting methods, etc	4	[37], [70], [86], [158]
Temporal calibration & Scoring	platt calibration, circadian weighting, etc	3	[143], [161], [163]
Meta-decision framework	LTT	1	[137]

The voting mechanism is the most widely used post-processing strategy, appearing in 31 studies. Among these approaches, the k-of-n scheme predominates and is reported in 15 studies. In this approach, a window of n consecutive segments is examined, and an alarm is issued once at least k segments within that window are classified as positive. The segment-ratio method—issuing an alarm when the proportion of positive segments in the window exceeds a preset threshold—appears in eight studies, whereas the consecutive—segment method—requiring several successive positive segments to confirm an alarm—is employed in six studies. Less common variants include the segment-max approach, which triggers an alarm based on the maximum probability (or score) among segments in the window and is reported by only one study [172], as well as the multi-channel voting method, which integrates votes across multiple EEG channels and is also documented in a single study [129].

The threshold mechanism is reported in 22 studies, with fixed thresholding representing the dominant method, adopted by 21 studies. In this approach, a threshold is predefined, and an alarm is triggered once the model output exceeds this threshold. By contrast, adaptive thresholding, which dynamically adjusts the threshold based on changing conditions or patient-specific data, is rarely explored, having been reported in only a single study [162].

TABLE 14. Top three most frequently used combination postprocessing methods.

Combination Method	Freq.	Ref.
k-of-n	15	[28], [40], [41], [46], [50], [56], [85], [90], [93], [112], [116], [125], [134], [135], [149]
Moving-average smoothing + thresholding	5	[57], [95], [99], [104], [144]
Moving-average smoothing + thresholding + 30 min refractory period	4	[51], [96], [140], [160]

To curb prediction volatility and reduce false-positive alarms, many studies apply temporal smoothing and debouncing techniques. Moving-average smoothing is the most common, appearing in 21 studies; it averages prediction scores or probabilities over a sliding window to produce steadier outputs. A refractory-period scheme is reported in 10 studies, inserting a fixed silent interval after each alarm to prevent rapid re-triggering. Finally, a sliding-window counting approach, used in five studies, tallies positive detections within a preset interval and issues an alarm only when the count exceeds a predefined threshold.

Only a handful of studies investigate advanced fusion strategies designed to boost generalization and sustain stable, long-term predictions. Four studies discuss adaptive

TABLE 15. Overview of post-hoc explainability mechanisms.

Category	Representative Methods	Freq.	Ref.
Feature attribution / contribution scoring	LRP, SHAP, LIME	4	[25], [29], [142], [160]
Statistical / network visual analytics	Functional connectivity networks (directed transfer function, functional connectivity), KL divergence, feature-importance ranking	4	[65], [109], [115], [147]
Gradient-based activation visualisation	Grad-CAM, Saliency map	3	[104], [128], [165]

ensemble mechanisms, introducing advanced fusion approaches such as MAML-based ensemble classifiers [86], simple fully connected ensemble classifiers [158], and multi-classifier voting methods [37], [70]. Temporal calibration and scoring techniques are reported in three studies, involving specific approaches like platt calibration [163], circadian weighting which accounts for circadian rhythms [161], and both chronological and cumulative firing power methods [143], all of which integrate prediction probabilities over time. Lastly, the meta-decision framework is introduced in only one study through the learn-then-test (LTT) approach [137], dynamically adjusting predictions following an initial learning phase to improve statistical robustness.

However, as Table 14 illustrates, combination methods are the most frequently employed postprocessing strategies in these studies.

The k-of-n voting scheme is the most prevalent strategy, appearing in 15 studies. The next most common workflow—reported in 5 studies—first smooths model probabilities using a moving-average filter and then applies a fixed threshold, thereby attenuating transient spikes and stabilizing the decision boundary. An additional refinement, documented in four studies, appends a 30-minute refractory period to this pipeline, such that once an alert is issued, any subsequent detections occurring within the refractory window are automatically suppressed.

Although front-end networks continue to grow deeper, the final alert logic remains predominantly heuristic. Approximately one-third of the studies implement a simple k-of-n voting scheme or a fixed probability threshold. Another fifth incorporate moving-average smoothing or a refractory silent period; however, adaptive or risk-aware calibration remains largely absent. Only four studies explore learned ensembles or meta-decision layers, and just one introduces dynamic thresholds aligned with circadian context. This continued reliance on hard-coded postprocessing leads many systems to exhibit elevated false-alarm rates under conditions of degraded signal quality or shifting patient states—constituting a major barrier to real-world deployment.

F. CLINICAL INTERPRETABILITY AND DEPLOYMENT

To bridge the gap between algorithmic development and real-world use, this section examines two complementary facets of model practicality: first, the post-hoc interpretability techniques that make predictions transparent, and then the actual deployment strategies that bring these models into clinical or edge environments.

1) INTERPRETABILITY

To assess how interpretability is addressed in practice, we cataloged the post-hoc explainability techniques reported across the literature. Of the 149 seizure prediction studies reviewed, only 11 explicitly integrate an explainability module, which highlights a considerable gap between algorithmic performance and clinical transparency. As summarized in Table 15, the post-hoc explainability mechanisms employed in EEG-based seizure prediction studies predominantly fall into three main categories: feature attribution or contribution scoring, statistical or network visual analytics, and gradient-based activation visualisation.

Feature-attribution methods, reported in four studies, mainly utilize algorithms such as layer-wise relevance propagation (LRP), SHapley additive exPlanations (SHAP), and Local Interpretable Model-Agnostic Explanations (LIME), generating heat maps that highlight the relevance of EEG channels or frequency bands. For example, Jemal et al. [25] applied LRP to their CNN by backpropagating relevance scores from the output back to each input channel and time point, producing heat-maps that directly visualise which EEG channels drove the model's seizure prediction decisions.

Meanwhile, statistical or network visual-analytics approaches, reported in four studies, leverage functional-connectivity analyses (e.g., directed transfer function), Kullback–Leibler (KL) divergence, and feature-importance ranking to interpret model decisions by quantifying changes in statistical patterns or network structures across seizure states. For example, Wang et al. [65] computed directed transfer function–based information-flow features between iEEG channels, rebuilt them into channel-frequency maps, and then used BrainNet Viewer to plot the resulting functional networks—clearly showing how preictal connectivity patterns differ from interictal ones.

Finally, gradient-based activation-visualization techniques, reported in three studies, apply methods such as gradient-weighted class activation mapping (Grad-CAM) and saliency maps to highlight the regions of spectro-temporal EEG representations on which the model focuses. For example, Guo et al. [104] used Grad-CAM on their STS-Net's pyramid convolution outputs—back-propagating class gradients to each rhythm's feature map—to produce heat-maps that pinpointed δ and θ band activations as the most discriminative for preictal prediction.

TABLE 16. Overview of deployment categories for EEG-based seizure prediction systems.

Category	Description	Ref.
Hardware-accelerated deployment	Low-power, real-time inference implemented on dedicated FPGA/ASIC hardware	[127], [152]
IoT edge, cloud, LTE deployment & performance evaluation	Integrated on LTE edge nodes or consumer-grade IoT platforms, with real-time performance reports	[64], [141], [165]
Conceptual/architectural scheme	System or network architecture proposed without implementation or experimental data	[63], [77], [114]
Clinical prototype validation	Hardware prototype tested in real patients or controlled hospital environments	[167]

2) DEPLOYMENT

To evaluate how research prototypes have been translated into practical systems, we analyze the deployment approaches employed in EEG-based seizure prediction studies.

Currently, the majority of studies focus on developing and training prediction models without explicit deployment or practical validation. Typical computational setups reported in these training-focused studies exhibit significant diversity. For instance, Truong et al. [112] trained their model in a Python 2.7 environment using Keras 2.0 with a TensorFlow 1.4.0 backend. They parallelized convolutional neural network computations across four NVIDIA Tesla K80 GPUs (each 24 GB VRAM; 96 GB total), markedly accelerating the processing of 30 s short-time Fourier transform (STFT) feature matrices. Toraman et al. [67] ran experiments in Python 3.5 with Keras 2.1.6 on a Linux server equipped with an NVIDIA GTX 1080 GPU and at least 16 GB of system RAM, enabling efficient forward and backward propagation on 224 $\times$ 224-pixel EEG spectrograms. Singh et al. [63] trained their CNN on a Windows 10 workstation featuring an Intel Core i7-8750H CPU (2.21 GHz), 16 GB RAM, and an NVIDIA GeForce GTX 1060 Max-Q GPU (6 GB VRAM; 1.506 GHz), under Python 3.8, supporting end-to-end preprocessing and training workflows. Dong et al. [116] performed training on a workstation with an AMD Ryzen 5 3400G CPU (3.70 GHz), 32 GB RAM, and an NVIDIA GeForce RTX 2080 GPU, using Python 3.8, PyTorch 1.12.1, and MATLAB R2020a. Saadoon et al. [37] utilized MATLAB's deep learning toolbox on a workstation with an Intel Core i9 processor, 32 GB DDR4 RAM, and an NVIDIA RTX 3090 GPU (24 GB VRAM) for EfficientNet-B0 feature extraction and a six-model SVM ensemble classification. These heterogeneous computational resources and framework selections indicate that most studies remain at the prototype stage and are still a considerable distance from practical system deployment.

Despite the extensive detailing of computational environments for model development, only nine studies have progressed to actual system deployment. As summarized in Table 16, 2 studies demonstrated hardware-accelerated inference leveraging field-programmable gate array (FPGA) or application-specific integrated circuit (ASIC) technology to achieve low-power, high-speed processing essential for real-time seizure prediction. Khalil et al. [127] implemented the proposed convolutional autoEncoder–ensemble long short-term memory (CAE-eLSTM) hardware predictor in stand-alone VHDL on an Altera Arria 10 GX FPGA and synthesized it to 45 nm ASIC technology using Synopsys Design Compiler at 120 MHz, achieving 1.53 mW power consumption. Massoud et al. [152] deployed the patient-specific MLP seizure prediction model on a Xilinx Virtex-7 FPGA using 10-bit fixed-point arithmetic, occupying 65401 LUTs and 10208 registers with 7.3% device utilization, and achieving 200 MHz operation at 1.5 W power—demonstrating markedly higher energy efficiency than inference on an NVIDIA RTX 2060 GPU.

Additionally, three studies respectively deploy seizure prediction models or conduct performance testing on an IoT edge device, a cloud platform, and an LTE edge-computing testbed. Ahmad et al. [64] trained and evaluated their CNN–BiLSTM attention-based multi-feature fusion model on a Lenovo Legion PC with 2.60 GHz Intel Core i9 CPU, 32 GB RAM and 8 GB NVIDIA RTX 2060 Max-Q GPU using Python 3.7, TensorFlow and Keras and deployed the CIoT framework on a Raspberry Pi 4B with 8 GB LPDDR4 SDRAM, dual-band Wi-Fi, Bluetooth 5.0 and Gigabit Ethernet for real-time EEG acquisition and SMS alerts via AWS IoT. Hassoon et al. [141] deployed a cloud-based, real-time EEG signal acquisition, preprocessing, and CAE-CNN inference pipeline on AWS elastic compute instances using a three-tier infrastructure architecture. Hosseini et al. [165] implemented an LTE edge-computing testbed using USRP B210 software-defined radios with the OpenAirInterface LTE stack on a virtual machine and conducted performance testing.

Meanwhile, three studies propose conceptual system architectures, outlining potential deployment frameworks, data flow, and interaction schemes between model components and external hardware, although these lacked experimental validation or practical implementation details. Finally, just one study present a tangible clinical prototype explicitly tested within a hospital or patient setting, showcasing direct clinical interaction, usability assessment, and preliminary validation under realistic usage scenarios [167].

Only 11 of 149 papers incorporate any post-hoc explainability—typically LRP, SHAP, or Grad-CAM—and fewer than ten present hardware, edge, or IoT prototypes. A single study reports bedside testing with real patients.

Without transparent rationales and on-device feasibility data, clinicians remain justifiably sceptical of "black-box" predictions, and regulatory pathways stay unclear. Bridging this gap will require embedding explainability into user interfaces, validating models under hospital workflows, and demonstrating low-power, low-latency inference on wearable or implantable platforms.

IV. DISCUSSION

While this review synthesizes major challenges and promising directions across five core dimensions of seizure prediction research datasets, preprocessing methods, model architecture, validation strategy, and decision refinement, but it is limited by (i) searches confined to *PubMed* and *Web of Science* and (ii) the inclusion of English-language studies only, which may have excluded relevant non-indexed or non-English work.

A. DATASETS

Statistical profiling of 149 primary studies reveals three structural imbalances in the dataset landscape that fundamentally shape what today's "state of the art" actually measures. First, a pronounced pediatric sEEG bias dominates current practice: 67.1% of studies rely exclusively on public datasets, with the CHB-MIT dataset alone accounting for 71.8% of all dataset citations. Since CHB-MIT comprises recordings from only 23 children, sampled at 256 Hz using a 23-channel montage, models achieving high performance on this benchmark are more likely leveraging a narrow, age-specific feature space than demonstrating broad clinical robustness. Pediatric EEG signals characteristically exhibit stronger delta activity and distinct artifact profiles compared to adult EEG. As a result, filters trained on this dataset tend to overemphasize low-frequency components while underrepresenting fast rhythms critical for detecting seizure onset in adults, leading to measurable drops in sensitivity when generalized to adult cohorts.

Second, long-term, high-density iEEG remains chronically underutilized. Private repositories such as EPILEPSIAE (nine studies) and Freiburg (eight studies) provide up to 122 channels at 1 kHz sampling rates, yet collectively appear in fewer than 12% of publications. Intracranial recordings capture high-frequency oscillations (80 to 250 Hz) and sub-second phase couplings that are not observable in scalp EEG. These features enable earlier and more localized seizure warnings, but are rarely learned by current algorithms. Moreover, iEEG datasets often span multiple weeks of continuous monitoring, revealing circadian and multiday seizure cycles absent in scalp benchmarks. Consequently, biomarkers that depend on high-frequency activity or fine-grained spatial synchrony remain validated on limited samples, constraining the generalizability of localization studies.

Third, genuine cross-source integration remains rare. Only 13.4% of studies combine public and private datasets——typically via a "pre-train on CHB-MIT, fine-tune on local data" paradigm. However, in the absence of standardized cross-dataset evaluation protocols, reported transfer gains are difficult to interpret. Research groups often employ custom data splits, variable resampling strategies, and inconsistent channel selections; thus, a reported 5% AUC improvement in one study may correspond to a 10% drop under another lab's configuration. Without harmonized inter-database benchmarks, it remains impossible to disentangle architectural innovation from favorable domain alignment.

In sum, many reported performance gains likely reflect overfitting to a pediatric scalp EEG niche rather than genuine advances toward generalizable, bedside seizure prediction systems. These issues stem from substantial dataset imbalances, including overreliance on pediatric data and limited adoption of high-density intracranial recordings. To improve robustness across diverse clinical scenarios, future research should prioritize the integration of more representative EEG datasets——encompassing adult populations, multiple recording modalities, and a broader spectrum of clinical use cases.

B. PREPROCESSING

Among the 149 studies surveyed, a clear "convenience paradigm" emerges in EEG preprocessing. Of the studies that explicitly report segment duration, 65.6% adopt epochs of 10 s or shorter. The most commonly used preictal window falls within the 30–60-min range, and approximately 58.4% of studies rely solely on standard notch or high-/low-pass filters for signal denoising. Class imbalance is most often addressed through interictal undersampling, reported in 57 studies, while only 19 apply sliding-window augmentation, and fewer than 10 use advanced techniques such as SMOTE or GAN-based synthesis. These streamlined preprocessing choices improve code portability and increase data volume, facilitating the reproduction of early CNN- and LSTM-based baselines. However, they introduce three critical methodological constraints.

First, segmenting EEG into epochs shorter than 10 seconds implicitly assumes that all discriminative preictal cues are strictly local. This practice truncates hour-scale dynamics—such as gradual shifts in dominant frequency or multiday circadian rhythms—thereby limiting the model's access to broader temporal context and reducing the achievable warning horizon. Second, the widespread use of raw or lightly filtered EEG shifts the burden of artifact rejection onto the model. While deep networks often learn to suppress line noise and ocular artifacts under controlled conditions, they may also attenuate subtle high-frequency oscillations (> 80 Hz) that serve as important intracranial biomarkers, thus diminishing predictive performance. Third, class imbalance is most commonly addressed by discarding interictal data. With 57 studies undersampling the majority class, models train on seizure-rich data streams that poorly reflect the distribution of real-world EEG. When deployed in continuous bedside settings, such models encounter extended periods of background activity absent during training, which often results in false-alarm rates that exceed clinically acceptable thresholds, despite reporting high cross-validated accuracy in laboratory settings.

As a result, the field continues to rely on one-size-fits-all preprocessing choices that limit comparability and generalizability. To overcome these limitations, future research systematically explores the effects of varying epoch lengths and prediction horizons across multiple temporal scales. It also rigorously compares state-of-the-art denoising algorithms with raw-signal pipelines. Equally important is the adoption of robust class-balancing methods—such as generative augmentation or active resampling—which provide richer and more realistic alternatives to simple undersampling.

C. MODEL STRUCTURE

Our review reveals that contemporary EEG-based seizure prediction adheres to a ''convolution-dominated, end-to-end-emerging, heavy time–frequency-reliant'' paradigm. Approximately 71.8% of studies employ pure deep learning architectures—predominantly CNN, followed by CNN–RNN cascades, CNN–Transformer hybrids, and a small number of pure Transformer models—while promising alternatives such as GNN, TCN, and VAE remain limited to isolated proofs of concept. Within these pipelines, dimensionality reduction is almost always implicit: 83% of studies describing their approach rely on reductions built into the architecture rather than on explicit feature selection steps, further narrowing the representational scope to local time–frequency textures and leaving phase synchrony or cross-channel graph structure largely unexplored. Compounding this methodological narrowness, only a small minority of studies benchmark against basic linear baselines (e.g., LDA, LR, linear SVM), making it difficult to gauge the true incremental value of convolution-heavy frameworks and risking overestimation of ''state-of-the-art'' gains.

First, this skew largely reflects the practicality of CNN on small, well-curated datasets such as CHB–MIT, where local time–frequency textures dominate and annotated data are limited. Most pipelines either discard handcrafted descriptors entirely or first project the raw EEG waveform into two-dimensional time–frequency representations before model training begins. Although this strategy accelerates convergence, it also removes phase information and inter-channel relationships, both of which are crucial for modeling the network dynamics that precede a seizure. Moreover, it tends to couple the learned filters tightly to the noise characteristics and sampling scheme of a specific dataset, thereby limiting generalization to larger and more heterogeneous dataset such as Temple University Hospital EEG Corpus (TUH). Until the field systematically benchmarks raw waveform-based, graph-aware, and Transformer-based models under unified cross-patient evaluation protocols, this convolution-heavy paradigm continues to produce impressive in-lab accuracies that do not translate reliably into patient-specific forecasts in real-world settings.

Second, because dimensionality reduction is embedded within hidden mechanisms such as convolutional strides, pooling layers, and automated channel mixing, it operates as a virtual ''black box'': researchers have no direct control over which channels, frequency bands, or temporal scales survive compression and in what proportions. This opacity not only erodes interpretability and clinician trust but also weakens reproducibility and precludes systematic, principled exploration of alternative compression strategies that might better preserve patient-specific dynamics.

Third, a pervasive limitation is the widespread exclusion of basic linear baselines. Researchers often assume that deep learning architectures inherently outperform simpler models, leading to the omission of linear approaches as trivial or already well established—particularly given novelty expectations and publication space constraints. Incentives within the publication process further exacerbate this bias, favoring reports of large performance gains rather than more tempered but rigorous comparisons to modest baselines. In addition, linear models often require deliberate adaptation to fit within deep learning-oriented preprocessing pipelines, which further reduces their appeal. Without such comparisons, it remains difficult to disentangle genuine modeling advances from artifacts introduced through extensive preprocessing or dataset-specific tuning. Incorporating well-tuned linear baselines substantially enhances the interpretability and credibility of performance claims for complex deep learning models.

Consequently, the field remains dominated by convolution-centric architectures and relies primarily on implicit dimensionality reduction, yet only a limited fraction of studies compare these models with basic linear predictors. Future research should establish standardized benchmarks that consistently include linear baselines, evaluate CNN, RNN, Transformer, and emerging architectures such as GNN and TCN under unified experimental settings, and explore lightweight hybrid models that integrate attention or graph-based components into compact backbones to balance accuracy and efficiency. Furthermore, a unified evaluation framework that considers data dimensionality, sample size, real-time constraints, and interpretability goals is essential when assessing implicit, explicit, or hybrid dimensionality reduction strategies.

D. EXPERIMENTAL VALIDATION

Experimental validation in seizure prediction research consistently relies on a narrow set of default configurations. Approximately one-quarter of studies set the SOP at 30 min, while another fifth adopt a 5-min SPH. Sensitivity and accuracy dominate the performance metrics, whereas clinically meaningful measures—such as the FPR/h—appear in fewer than half of the publications, and time-in-warning is reported only occasionally. Validation strategies also remain limited, with most studies relying on single hold-out splits or leave-one-seizure-out protocols; cross-patient evaluation using leave-one-patient-out is applied in only a small minority of cases. These preferences often reflect practical constraints:

the 30-min/5-min window aligns with clinical routines and regulatory expectations; sensitivity and accuracy are intuitive and widely accepted; and simple data partitioning methods are easier to implement, particularly in early-stage experiments or when datasets are limited.

Although these defaults support reproducibility and facilitate direct comparisons, they also introduce several critical limitations. First, the use of a fixed 30-min SOP and 5-min SPH implicitly assumes a uniform prodromal duration and intervention window, overlooking patients whose seizures are preceded by only seconds of warning or who require extended lead time for treatment or caregiving logistics. Second, segment-level sensitivity and accuracy may obscure the risk of alarm fatigue: a model that issues warnings every 10 min may achieve 99% sensitivity on imbalanced test sets, yet remain clinically impractical due to excessive false alarms. Third, single hold-out validation fails to account for domain shifts caused by differences in EEG montages, recording artifacts, or acquisition hardware, often resulting in inflated performance estimates that do not generalize to new clinical settings or wearable platforms. Finally, the lack of uncertainty quantification and cost-sensitive metrics limits the adaptability of models to diverse clinical use cases, such as pediatric versus adult care or closed-loop stimulation versus caregiver notification.

Continued reliance on fixed SOP/SPH configurations, limited performance metrics, and simplistic validation protocols inflates reported accuracy, conceals the risk of alarm fatigue, and weakens confidence in the generalizability of seizure prediction systems. To overcome these limitations, future work prioritizes the use of variable, patient-specific windows; incorporates clinically meaningful evaluation criteria—such as FPR/h, time-in-warning, and calibrated probability estimates; and validates models across patients, devices, and clinical environments to reflect real-world variability. Adopting leave-one-patient-out or external-dataset testing, integrating uncertainty estimates, and involving clinicians early in defining acceptable false-alarm thresholds further supports the translation of promising laboratory models into reliable bedside seizure-warning solutions.

E. DECISION REFINEMENT & CLINICAL TRANSLATION

Most epileptic seizure prediction pipelines continue to treat the decision-refinement layer as an ancillary component rather than an integral part of the model. The dominant practices—k-of-n majority voting, fixed confidence thresholds, and simple moving-average filters—offer ease of implementation but reduce to coarse, manual tuning of raw prediction scores. These heuristic approaches disregard the full probability distribution of model outputs, neglect temporal context beyond short windows, and lack the flexibility to adapt to baseline shifts induced by medication changes, circadian rhythms, or electrode displacement. Only a limited number of studies explore more adaptive postprocessing strategies, such as combining multiple techniques or applying dynamic thresholds. However, these efforts remain confined to proof-of-concept experiments on static datasets and fail to address the challenges of generalization when models are deployed across different electrode montages, wearable sensors, or out-of-hospital environments.

Interpretability further compounds this issue. Despite its clinical relevance, only 11 studies apply explanation techniques such as LIME, SHAP, Integrated Gradients, or Grad-CAM to identify which channels, frequency bands, or time segments contribute to model decisions. Without transparent interpretive feedback, clinicians are left unable to assess whether false alarms stem from artifacts, feature biases, or true preictal signals, which undermines confidence in the system's reliability.

Translation from proof-of-concept to functional deployment on point-of-care or wearable platforms also remains rare. Few studies progress beyond offline evaluation to on-device inference, and even fewer conduct prospective real-time testing in clinical settings. This gap arises from several interrelated challenges: deep learning models often exceed the computational constraints of portable hardware; patient-specific variability necessitates continuous model adaptation, which is difficult to implement on embedded systems; and integration with hospital information systems or caregiver alert mechanisms introduces additional concerns around cybersecurity, data privacy, and interoperability.

The ongoing dependence on heuristic postprocessing, minimal use of interpretability tools, and lack of real-world deployment collectively hinder generalizability, reduce clinician trust, and prevent most seizure prediction systems from advancing beyond in-silico prototypes into reliable, clinically deployable early-warning solutions.

V. CONCLUSION

This work presents the first quantitative survey of 149 EEG-based seizure prediction studies published between 2018 and May 2025. For each study, the datasets employed, preprocessing procedures, model architectures, validation protocols, postprocessing strategies, and any claims related to interpretability or deployment are systematically documented, thereby encompassing the entire pipeline from data acquisition to potential bedside application. The search methodology and screening criteria are described in a step-by-step manner to facilitate reproducibility and allow other researchers to replicate or extend the corpus.

Based on this analysis, four actionable contributions are identified: these collectively characterize prevailing trends and methodological advances in the field (Objective (i)) while also highlighting the persistent limitations that continue to hinder progress in seizure prediction research (Objective (ii)).

- 72.8% of dataset citations concentrate on the pediatric CHB-MIT sEEG dataset, while fewer than 10% utilize high-density iEEG. This imbalance leads to overfitting on short, low-channel pediatric recordings and limits generalization to adult populations, high-channel configurations, and long-term monitoring scenarios, thereby constraining clinical translation.

- The prevailing convenience paradigm—characterized by sub 10-s epochs, a 30 to 60 min preictal window, routine reliance on basic notch or band-pass filtering for signal denoising, and interictal undersampling—facilitates experimentation and inflates in lab accuracy. However, it also truncates long-range temporal dependencies, induces a distribution shift from training to deployment, and obscures the nature of lost predictive information, thereby undermining both interpretability and real-world robustness.
- Model development remains dominated by convolution- and time-frequency-based architectures, while exploration of more flexible alternatives such as GNN and Transformer is still sparse. Dimensionality reduction is handled almost entirely implicitly, leaving researchers without explicit control over the information retained. Moreover, only a minority of studies benchmark their models against simple linear baselines, making it difficult to determine whether reported performance gains stem from genuine modeling advances or from added architectural complexity and preprocessing effort.
- Experimental protocols disproportionately prioritize convenience—favoring fixed 30-min SOP and 5-min SPH windows, as well as simplistic validation strategies such as single hold-out or leave-one-seizure-out splits—over clinical realism. Postprocessing commonly relies on static, opaque heuristics such as fixed thresholds and k-of-n voting, offering limited interpretability. Translation from proof-of-concept to bedside or wearable deployment remains exceedingly rare. As a result, models often fail to generalize, produce poorly calibrated alerts with high false-alarm and missed-seizure rates, and erode clinician trust, ultimately hindering real-world adoption.

To address these limitations (Objective (iii)), we recommend that future research:

- Assemble diverse and representative EEG datasets that encompass both pediatric and adult populations, integrate high-density intracranial alongside scalp recordings, and capture a broad spectrum of clinical scenarios. This approach mitigates the field's current overreliance on the CHB-MIT benchmark and supports more robust generalization.
- Replace convenience-driven preprocessing with adaptive, multi-scale pipelines by evaluating varied epoch lengths and prediction horizons, benchmarking denoising techniques under realistic deployment constraints, and addressing class imbalance through generative augmentation or active resampling instead of simple undersampling.
- Establish unified benchmarks that systematically pit CNN, RNN, Transformer, GNN, TCN, and other emerging architectures against well-tuned linear baselines, thereby propelling flexible, non-convolutional designs beyond the prevailing time–frequency paradigm and mandating ablation experiments that contrast default implicit compression with explicit, task-aware dimensionality reduction. This allows for meaningful comparisons that isolate true modeling improvements while balancing predictive accuracy, computational efficiency, and interpretability.
- Adopt clinically aligned evaluation protocols that incorporate patient-specific splits, clinically meaningful metrics (e.g., FPR/h, time-in-warning), and both cross-patient and external-dataset validation. In addition, static postprocessing thresholds should be replaced with dynamic, risk-aware mechanisms. Interpretability should be embedded by design, and final systems should be deployed on energy-efficient edge hardware with prospective validation in both clinical and home settings.

Only through comprehensive reform can seizure prediction models be transformed from promising prototypes into reliable solutions for real-world clinical application.

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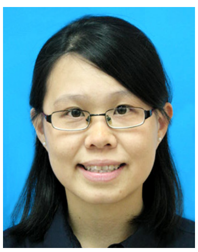

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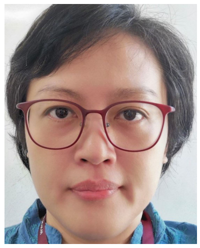

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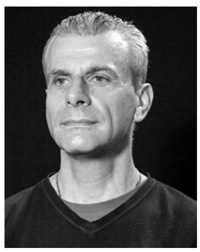

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